Unit 202 — Final Visual Production Review

Package: CC-11.7B · Commit: `038f97356f080dc9b60159814d46489254fcbdca` · Generated: 2026-08-24T00:25:44.870Z

PRE-PRODUCTION REVIEW — NO ARTWORK GENERATION SHOULD BEGIN UNTIL APPROVED

This document is the complete, final pre-production review pack for the Unit 202 instructional-visual production catalogue. It is generated directly from the live Visual Production Studio catalogue -- every number and table in this document is derived mechanically, never hand-maintained separately.

It is intended to be understandable without opening the repository: an independent reviewer should be able to answer, from this document alone, whether every important visual need is represented, whether any distinct images have been incorrectly compressed into one production asset, whether teaching/assessment/feedback states are properly separated, and whether the proposed manual ChatGPT production workload is sensible.

Executive summary

Visual families	21
Production assets	53
Canonical learner-visible states	98
Historical CC-05D variants reconciled	66 / 66
REQUIRED assets (total / ready / blocked)	42 / 24 / 18
USEFUL assets (total / ready / blocked)	10 / 2 / 8
Deterministic-only assets (no ChatGPT job)	11
Premium/hybrid art jobs — TOTAL	42
of which REQUIRED	34
of which USEFUL	8
Shared-base decisions audited	30
KEPT (SAFE_SHARED_BASE)	18
KEPT WITH CONDITIONS	3
SPLIT (SEPARATE_ARTWORK_REQUIRED)	9
REQUIRED VISUAL PRODUCTION COMPLETE	NO — not yet complete

How to read this catalogue

VisualFamily	An organisational grouping of one or more ProductionAssets that together teach one concept. Never reduces prompt granularity.
ProductionAsset ("image job")	One real, independent image-generation job (or deterministic-only artefact) -- its own prompt, filename, save slot and approval state.
CanonicalState	One distinct learner-visible presentation a ProductionAsset supports (e.g. teaching vs assessment). Several states may safely share one base image via deterministic overlay, or may require separate artwork -- see the Multi-state sharing review.
Deterministic asset	Produced entirely by ALP's own rendering code -- never a ChatGPT image-generation job.
Premium / Hybrid asset	A real ChatGPT image-generation job. Hybrid assets combine generated artwork with a deterministic overlay layer.
Required	Unit 202 visual completeness depends on this asset existing.
Useful	Optional enrichment the Product Owner may choose to commission for course quality -- never blocks REQUIRED completeness.
Blocked reference	A production-readiness state, independent of REQUIRED/USEFUL: no approved reference exists yet for this asset.
Teaching / Assessment / Feedback	The pedagogical state a canonical state is produced for -- assessment states never reveal the answer; teaching/feedback states may.

Full catalogue, by LO / lesson

Every production asset in the live catalogue, 53 total.

LO1 — no current lesson (integration deferred, see reports/instructional-visuals/visual-needs-matrix.md)

#42 **Right-angle triangle / SOHCAHTOA** DEFERRED_SCOPE NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.trigonometry` · family: Right-angle triangle / SOHCAHTOA · LO1 — no current lesson (integration deferred, see reports/instructional-visuals/visual-needs-matrix.md) · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL

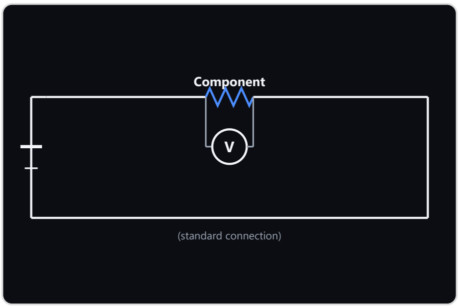
A right-angle triangle showing hypotenuse/opposite/adjacent relative to a selected acute angle, supporting SOHCAHTOA -- lesson integration remains deferred.

Reference	Standard right-triangle trigonometry reference -- to be selected when this asset is actually commissioned (to be recorded when selected)
Immutable facts	right angle present; hypotenuse opposite the right angle; opposite/adjacent sides correctly identified relative to the selected acute angle
Annotation policy	NONE
Canonical states	Right-angle triangle / SOHCAHTOA [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO2 — lesson.electrical.instrumentation

#17 **Ammeter / voltmeter / ohmmeter connections** REQUIRED NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.instrument.connections` · family: Instrument connections · LO2 — lesson.electrical.instrumentation · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL



A visually polished instrument-connection reference for style/contrast QA against the existing deterministic renderer -- connection correctness is already governed.

Reference	Wikimedia Commons — SimpleCircuit.SVG — https://commons.wikimedia.org/wiki/File:SimpleCircuit.SVG (recorded per source page)
Immutable facts	ammeter connects in series; voltmeter connects in parallel; resistance measurement requires a de-energised circuit; isolation/disconnection of an individual component is required only where necessary to avoid parallel-path readings, never claimed as a universal requirement; source/load context understandable
Annotation policy	NONE
Canonical states	voltmeter — parallel (standard) [MULTI_STATE] (historical-66) voltmeter — series (non-standard, deliberate teaching contrast) [MULTI_STATE] (historical-66) ammeter — series (standard) [MULTI_STATE] (historical-66) ammeter — parallel (non-standard, deliberate teaching contrast) [MULTI_STATE] (historical-66) ohmmeter — isolated (standard) [MULTI_STATE] (historical-66)
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: DETERMINISTIC_TECHNICAL -- instrument/connection-style states are deterministic vector topology, no generated artwork involved at all.
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#44 **Clamp meter recognition** USEFUL BLOCKED

`unit202.instrument.clamp-meter` · family: Instrument connections · LO2 — lesson.electrical.instrumentation · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

CC-11.7 audit finding: physical recognition of a clamp meter by its distinctive ferrite-jaw form -- genuinely different from the series/parallel/isolated connection topology the sibling TECHNICAL_DIAGRAM asset models. USEFUL enrichment, not REQUIRED.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	ferrite clamp jaw must be clearly visible and open-able around a conductor -- the defining recognition feature
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Clamp meter recognition (blocked) [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/instrument-clamp-meter-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#45 **Oscilloscope recognition** USEFUL BLOCKED

`unit202.instrument.oscilloscope` · family: Instrument connections · LO2 — lesson.electrical.instrumentation · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

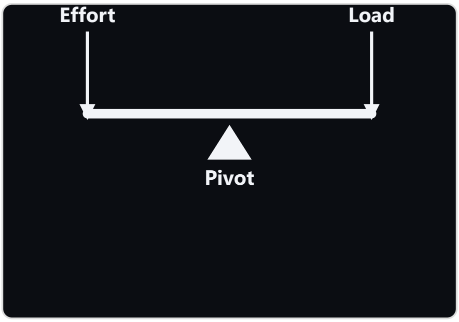
CC-11.7 audit finding: physical recognition of an oscilloscope by its distinctive screen/trace form. USEFUL enrichment, not REQUIRED.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	screen with a visible waveform trace must be clearly depicted -- the defining recognition feature
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Oscilloscope recognition (blocked) [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/instrument-oscilloscope-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO3 — lesson.foundation.physics.simple-machines

#07 **Lever — Class I** REQUIRED READY

unit202.levers.class-1 · family: Lever classes · LO3 — lesson.foundation.physics.simple-machines · role: CONFIGURATION · class: HYBRID

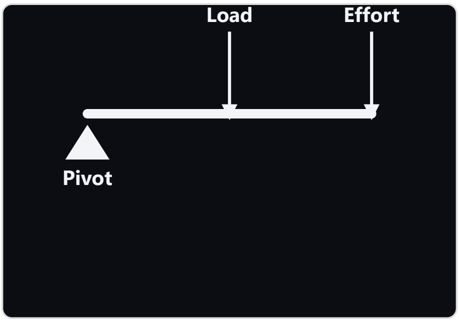


Show a Class I lever arrangement (fulcrum between effort and load) so a learner can recognise it from the arrangement itself.

Reference	Pearson Scott Foresman — Lever (PSF).svg — https://commons.wikimedia.org/wiki/File:Lever_(PSF).svg (Public domain)
Immutable facts	fulcrum positioned between the effort point and the load point
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Class I recognition (no distances) [MULTI_STATE] (historical-66) Class I with effort-arm/load-arm distances [MULTI_STATE] (historical-66)
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: 'recognition' and 'moment-balance' states depict the identical physical lever rig (same fulcrum/effort/load positions) -- the only difference is whether deterministic effort-arm/load-arm distance annotations are added on top. No pose, configuration or scene change.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/levers-class-1-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#08 **Lever — Class II** REQUIRED READY

unit202.levers.class-2 · family: Lever classes · LO3 — lesson.foundation.physics.simple-machines · role: CONFIGURATION · class: HYBRID

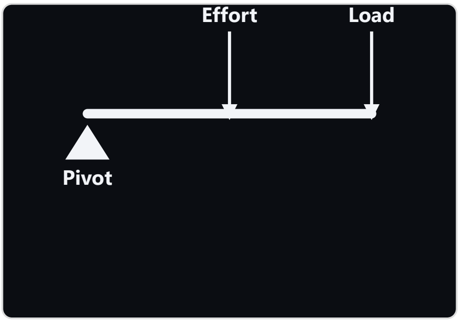


Show a Class II lever arrangement (load between fulcrum and effort) so a learner can recognise it from the arrangement itself.

Reference	Pearson Scott Foresman — Lever (PSF).svg — https://commons.wikimedia.org/wiki/File:Lever_(PSF).svg (Public domain)
Immutable facts	load positioned between the fulcrum and the effort point
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Class II recognition (no distances) [MULTI_STATE] (historical-66) Class II with effort-arm/load-arm distances [MULTI_STATE] (historical-66)
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: 'recognition' and 'moment-balance' states depict the identical physical lever rig (same fulcrum/effort/load positions) -- the only difference is whether deterministic effort-arm/load-arm distance annotations are added on top. No pose, configuration or scene change.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/levers-class-2-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#09 **Lever — Class III** REQUIRED READY

`unit202.levers.class-3` · family: Lever classes · LO3 — lesson.foundation.physics.simple-machines · role: CONFIGURATION · class: HYBRID

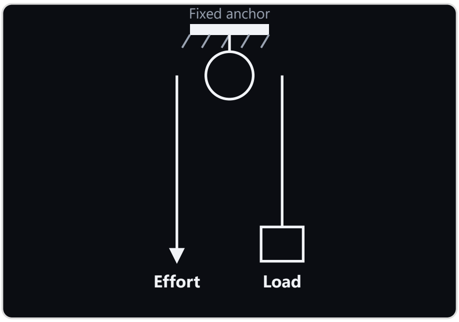


Show a Class III lever arrangement (effort between fulcrum and load) so a learner can recognise it from the arrangement itself.

Reference	Pearson Scott Foresman — Lever (PSF).svg — https://commons.wikimedia.org/wiki/File:Lever_(PSF).svg (Public domain)
Immutable facts	effort positioned between the fulcrum and the load point
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Class III recognition (no distances) [MULTI_STATE] (historical-66) Class III with effort-arm/load-arm distances [MULTI_STATE] (historical-66)
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: 'recognition' and 'moment-balance' states depict the identical physical lever rig (same fulcrum/effort/load positions) -- the only difference is whether deterministic effort-arm/load-arm distance annotations are added on top. No pose, configuration or scene change.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/levers-class-3-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#10 **Fixed pulley** REQUIRED READY

`unit202.pulleys.fixed` · family: Pulleys · LO3 — lesson.foundation.physics.simple-machines · role: CONFIGURATION · class: HYBRID



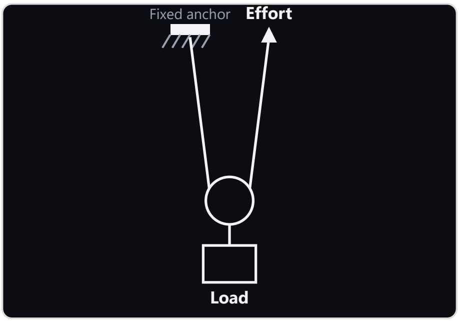
Show a fixed pulley: direction change only, mechanical advantage approximately 1.

Reference	Wikimedia Commons — Pulley1a.svg — https://commons.wikimedia.org/wiki/File:Pulley1a.svg (Public-domain where recorded)
Immutable facts	fixed anchor point; pulley wheel changes rope direction only; mechanical advantage approximately 1; physically continuous/plausible rope path
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Fixed pulley [MULTI_STATE] (historical-66)
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/pulleys-fixed-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#11 **Movable pulley**

REQUIRED READY

`unit202.pulleys.movable` · family: Pulleys · LO3 — lesson.foundation.physics.simple-machines · role: CONFIGURATION · class: HYBRID



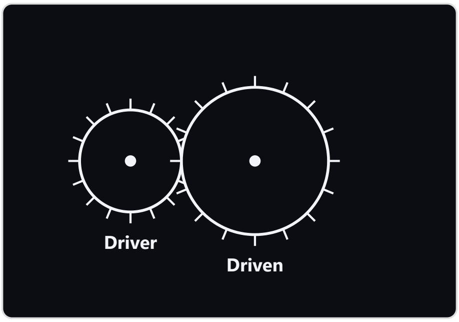
Show a simple movable pulley: two supporting rope segments, mechanical advantage approximately 2.

Reference	Wikimedia Commons — Pulley1a.svg — https://commons.wikimedia.org/wiki/File:Pulley1a.svg (Public-domain where recorded)
Immutable facts	exactly two supporting rope segments; mechanical advantage approximately 2; physically continuous/plausible rope path
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Movable pulley [MULTI_STATE] (historical-66)
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/pulleys-movable-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#18 **Driver/driven gears — driven gear larger than the driver**

REQUIRED READY

`unit202.gears.driven-larger` · family: Driver/driven gears · LO3 — lesson.foundation.physics.simple-machines · role: CONFIGURATION · class: HYBRID



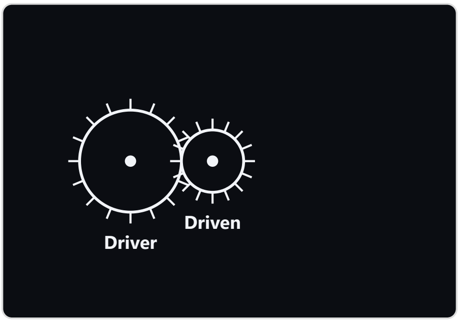
Show a driver gear meshed with a driven gear, driven gear larger than the driver, representing the gear ratio and the resulting torque/speed trade-off.

Reference	Wikimedia Commons — Example of a Compound Gear Train.png — https://commons.wikimedia.org/wiki/File:Example_of_a_Compound_Gear_Train.png (CC0)
Immutable facts	meaningful driver/driven relationship; physically plausible mesh; driven gear larger than the driver -- this specific size relationship is the defining physical fact of this asset, never mixed with a sibling size-ratio asset; correct rotation relationship when shown
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Driver/driven gears — driven larger [MULTI_STATE] (historical-66)
Shared-base decision	SEPARATE_ARTWORK_REQUIRED — CC-11.7B: split from the original unit202.gears (3 states on 1 asset). Driven-gear size relative to the driver is a genuine physical size change to a rendered object, not a labelling/overlay difference -- no deterministic overlay mechanism can resize a gear in generated artwork (only rotation-direction is declared as an overlay responsibility).
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/gears-driven-larger-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#51 **Driver/driven gears — driven gear smaller than the driver**

REQUIRED READY

`unit202.gears.driven-smaller` · family: Driver/driven gears · LO3 — lesson.foundation.physics.simple-machines · role: CONFIGURATION · class: HYBRID



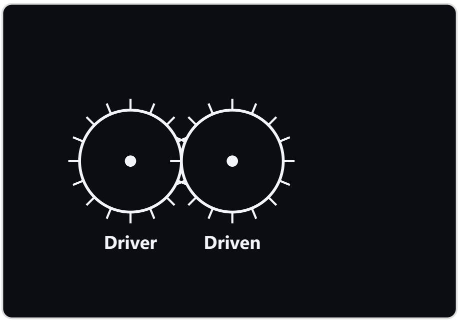
Show a driver gear meshed with a driven gear, driven gear smaller than the driver, representing the gear ratio and the resulting torque/speed trade-off.

Reference	Wikimedia Commons — Example of a Compound Gear Train.png — https://commons.wikimedia.org/wiki/File:Example_of_a_Compound_Gear_Train.png (CC0)
Immutable facts	meaningful driver/driven relationship; physically plausible mesh; driven gear smaller than the driver -- this specific size relationship is the defining physical fact of this asset, never mixed with a sibling size-ratio asset; correct rotation relationship when shown
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Driver/driven gears — driven smaller [MULTI_STATE] (historical-66)
Shared-base decision	SEPARATE_ARTWORK_REQUIRED — CC-11.7B: split from the original unit202.gears (3 states on 1 asset). Driven-gear size relative to the driver is a genuine physical size change to a rendered object, not a labelling/overlay difference -- no deterministic overlay mechanism can resize a gear in generated artwork (only rotation-direction is declared as an overlay responsibility).
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/gears-driven-smaller-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#52 **Driver/driven gears — driver and driven gears equal size**

REQUIRED READY

`unit202.gears.equal` · family: Driver/driven gears · LO3 — lesson.foundation.physics.simple-machines · role: CONFIGURATION · class: HYBRID



Show a driver gear meshed with a driven gear, driver and driven gears equal size, representing the gear ratio and the resulting torque/speed trade-off.

Reference	Wikimedia Commons — Example of a Compound Gear Train.png — https://commons.wikimedia.org/wiki/File:Example_of_a_Compound_Gear_Train.png (CC0)
Immutable facts	meaningful driver/driven relationship; physically plausible mesh; driver and driven gears equal size -- this specific size relationship is the defining physical fact of this asset, never mixed with a sibling size-ratio asset; correct rotation relationship when shown
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Driver/driven gears — equal [MULTI_STATE] (historical-66)
Shared-base decision	SEPARATE_ARTWORK_REQUIRED — CC-11.7B: split from the original unit202.gears (3 states on 1 asset). Driven-gear size relative to the driver is a genuine physical size change to a rendered object, not a labelling/overlay difference -- no deterministic overlay mechanism can resize a gear in generated artwork (only rotation-direction is declared as an overlay responsibility).
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/gears-equal-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#47 **Gear rotation-direction reversal / idler gear** USEFUL NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.gears.rotation-direction` · family: Driver/driven gears · LO3 — lesson.foundation.physics.simple-machines · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL

CC-11.7 audit finding: two meshed gears rotate in opposite directions; adding a third idler gear reverses the output direction back to match the driver without changing the overall ratio. Governed SUPPORTS-only content (non-mandatory).

Reference	Deterministic rotation-direction annotation on the sibling driver/driven gear reference geometry -- no separate photographic reference required (n/a -- deterministic annotation, not generated artwork)
Immutable facts	two directly meshed gears rotate in opposite directions; an idler gear between driver and driven reverses the output direction back to match the driver's own direction; an idler gear does not change the overall driver:driven ratio
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Two directly meshed gears — opposite rotation directions [TEACHING] Idler gear — output direction matches driver, ratio unchanged [TEACHING]
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: purely deterministic annotation (rotation-direction arrows, idler presence) on whichever sibling driver/driven gear base artwork applies -- no generated artwork involved at all, unaffected by the CC-11.7B size-ratio split.
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO4 — lesson.electrical.charge-and-current

#46 **Conventional current vs electron flow** USEFUL NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.current-direction.electron-flow-vs-conventional` · family: Conventional current vs electron flow · LO4 — lesson.electrical.charge-and-current · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL

CC-11.7 audit finding: a single wire with two labelled arrows -- conventional current (+ to -) and actual electron flow (- to +) -- shown together so the direction distinction and the underlying reason (electrons are negative, so they physically move opposite to the conventional-current convention) are both visible at a glance.

Reference	Standard conventional-current/electron-flow dual-arrow reference -- to be selected when this asset is commissioned (to be recorded when selected)
Immutable facts	conventional current direction: positive terminal to negative terminal; electron flow direction: negative terminal to positive terminal -- opposite to conventional current; both arrows on the same single wire/conductor, never on separate wires
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Conventional current vs electron flow (dual arrow) [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO4 — lesson.electrical.conductors-and-insulators

#40 **Conductor vs insulator** REQUIRED BLOCKED

`unit202.conductor-insulator` · family: Conductor vs insulator · LO4 — lesson.electrical.conductors-and-insulators · role: COMPARISON · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

Show a material-recognition comparison between conductors and insulators.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	—
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Conductor vs insulator (blocked) [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/conceptual/conductor-insulator-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO4 — lesson.electrical.fault-conditions-protection

#41 **Fuse vs circuit breaker comparison** REQUIRED BLOCKED

`unit202.protective-devices` · family: Fuse vs circuit breaker comparison · LO4 — lesson.electrical.fault-conditions-protection · role: COMPARISON · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

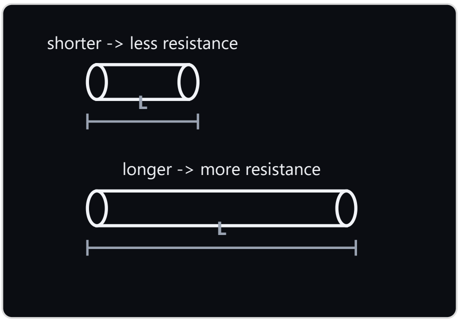
Show a fuse and a circuit breaker side by side, supporting cap.fault.compare_fuse_breaker (fuse must be replaced once blown; breaker can be reset), without endorsing one manufacturer's product appearance as canonical.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	one fuse and one circuit breaker shown side by side, not as two separate images; must not endorse one manufacturer's product appearance as canonical
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Fuse vs circuit breaker comparison (blocked) [TEACHING]
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B §5/§8 explicit pedagogical decision: the comparison itself (fuse must be replaced; breaker can be reset) is the teaching objective cap.fault.compare_fuse_breaker asks for, matching the brief's own 'ACCEPTABLE: Fuse vs circuit breaker comparison where comparison itself is the teaching objective' example. Confirmed as one intentional composite ProductionAsset, not two independent physical-recognition assets. ROLE corrected from the pre-audit PHYSICAL_RECOGNITION (a mismatch for a two-object comparison) to COMPARISON.
Expected output file	`apps/mobile/src/assets/instructional/unit202/conceptual/protective-devices-fuse-vs-breaker-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO4 — lesson.electrical.resistivity

#19 **Resistance vs conductor length** REQUIRED BLOCKED

`unit202.resistivity.length-comparison` · family: Resistance vs conductor dimensions · LO4 — lesson.electrical.resistivity · role: COMPARISON · class: HYBRID



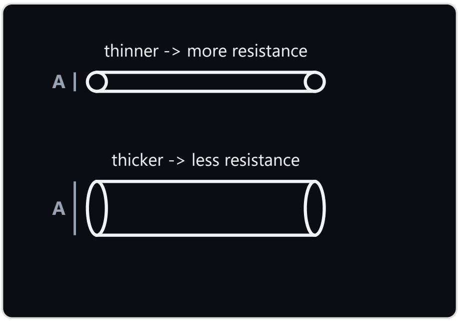
BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

Show two conductor rods differing only in length so a learner predicts the qualitative effect on resistance.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	increased length -> greater resistance (cross-sectional area unchanged)
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Resistance vs conductor length [MULTI_STATE] (historical-66)
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/resistivity-length-comparison-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#20 **Resistance vs conductor cross-sectional area** REQUIRED BLOCKED

`unit202.resistivity.area-comparison` · family: Resistance vs conductor dimensions · LO4 — lesson.electrical.resistivity · role: COMPARISON · class: HYBRID



BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

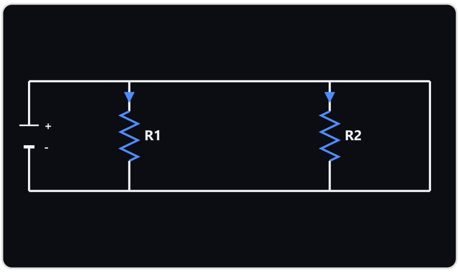
Show two conductor rods differing only in cross-sectional area so a learner predicts the qualitative effect on resistance.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	increased cross-sectional area -> lower resistance (length unchanged)
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Resistance vs conductor cross-sectional area [MULTI_STATE] (historical-66)
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/resistivity-area-comparison-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO4 — lesson.electrical.resistors-parallel

#15 **Parallel circuit** REQUIRED NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.circuit.parallel` · family: Parallel circuit · LO4 — lesson.electrical.resistors-parallel · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL



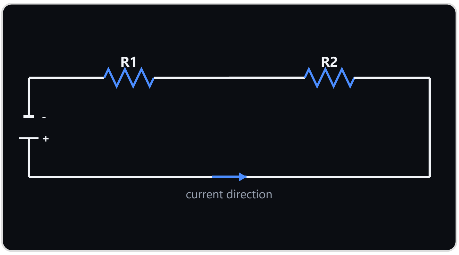
A visually polished parallel-circuit reference for style/contrast QA against the existing deterministic renderer -- topology and correctness are already governed.

Reference	Same circuit reference family as series (see unit202.circuit.series) — https://commons.wikimedia.org/wiki/File:Series_and_parallel_circuits2.svg (recorded per source page)
Immutable facts	source present; common pair of nodes/rails; separate branches; current direction consistent with source polarity
Annotation policy	NONE
Canonical states	Parallel circuit — 2 branches [MULTI_STATE] (historical-66) Parallel circuit — 3 branches [MULTI_STATE] (historical-66) Parallel circuit — 4 branches [MULTI_STATE] (historical-66)
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: DETERMINISTIC_TECHNICAL -- branch-count states are deterministic vector geometry, no generated artwork involved at all.
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO4 — lesson.electrical.resistors-series

#14 **Series circuit** REQUIRED NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.circuit.series` · family: Series circuit · LO4 — lesson.electrical.resistors-series · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL



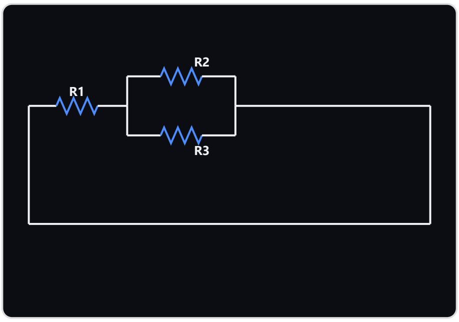
A visually polished series-circuit reference for style/contrast QA against the existing deterministic renderer -- topology and correctness are already governed.

Reference	Wikimedia Commons — Battery symbols and circuit.svg — https://commons.wikimedia.org/wiki/File:Battery_symbols_and_circuit.svg (recorded per source page)
Immutable facts	complete source; one continuous loop; UK/IEC component convention; current direction consistent with polarity when shown
Annotation policy	NONE
Canonical states	Series circuit — 2 components [MULTI_STATE] (historical-66) Series circuit — 3 components [MULTI_STATE] (historical-66) Series circuit — 4 components [MULTI_STATE] (historical-66)
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: DETERMINISTIC_TECHNICAL -- component-count states are deterministic vector geometry, no generated artwork involved at all.
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO4 — lesson.electrical.series-vs-parallel-comparison

#16 **Mixed series/parallel circuit** REQUIRED NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.circuit.mixed` · family: Mixed series/parallel circuit · LO4 — lesson.electrical.series-vs-parallel-comparison · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL



A visually polished mixed-topology reference for style/contrast QA against the existing deterministic renderer -- topology and correctness are already governed.

Reference	Wikimedia Commons — SeriesParallelCircuit.svg — https://commons.wikimedia.org/wiki/File:SeriesParallelCircuit.svg (use only as topology/reference, not a close stylistic derivative)
Immutable facts	genuinely mixed topology; obvious junctions; traceable current paths; source included where pedagogically necessary
Annotation policy	NONE
Canonical states	Mixed circuit — series of parallel [MULTI_STATE] (historical-66) Mixed circuit — parallel of series [MULTI_STATE] (historical-66)
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: DETERMINISTIC_TECHNICAL -- arrangement states are deterministic vector topology, no generated artwork involved at all.
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO4 — lesson.electrical.thermal-and-chemical-effects

#38 Chemical effect / electrolysis

REQUIRED READY

`unit202.electrolysis` · family: Chemical effect / electrolysis · LO4 — lesson.electrical.thermal-and-chemical-effects · role: PHENOMENON · class: HYBRID

ARTWORK NOT YET PRODUCED

Show the chemical effect of current: a source, an electrolyte and electrodes, with a meaningful current path -- no chemistry beyond syllabus scope.

Reference	Wikimedia Commons — Elektrolyse Allgemein.svg — https://commons.wikimedia.org/wiki/File:Elektrolyse_Allgemein.svg (prefer CC0/public-domain reference where available)
Immutable facts	source present; electrolyte present; electrodes present; meaningful current path
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Electrolysis cell — source, electrolyte, electrodes, current path [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/electrolysis-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#39 Heating effect of electric current

REQUIRED BLOCKED

`unit202.heating-effect` · family: Heating effect of electric current · LO4 — lesson.electrical.thermal-and-chemical-effects · role: PHENOMENON · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

Show the heating effect of electric current (resistive heating) at a conceptual level.

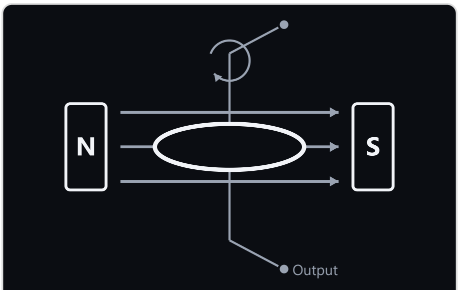
Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	—
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Heating effect of electric current (blocked) [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/conceptual/heating-effect-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO5 — lesson.emf.ac-generation-principles

#05 **Simple rotating-loop AC generator — loop plane facing poles — near-zero EMF**

REQUIRED READY

`unit202.generator.rotating-loop.horizontal` · family: Fleming's right-hand rule / AC generator effect · LO5 — lesson.emf.ac-generation-principles · role: PHENOMENON · class: HYBRID



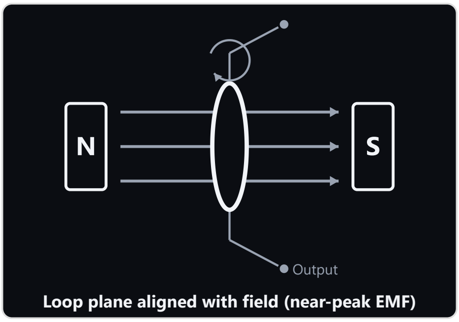
Loop plane facing poles (near-zero EMF)

Show a single loop of wire, loop plane facing the poles head-on (widest visible loop face), rotating on a central axis between N and S poles, establishing the physical basis of single-loop AC generation at Level 2 depth.

Reference	Wikimedia Commons — Diagram of single-phase generator with two poles.svg — https://commons.wikimedia.org/wiki/File:Diagram_of_single-phase_generator_with_two_poles.svg (CC0/public-domain reference material where recorded)
Immutable facts	N/S magnetic poles; loop/coil between the poles; loop plane facing the poles head-on (widest visible loop face) -- this specific pose is the defining physical fact of this asset, never mixed with the sibling pose asset; central rotational axis; output/slip-ring concept at governed Level-2 abstraction; rotating conductor cuts magnetic flux
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Loop plane facing poles — near-zero EMF [MULTI_STATE] (historical-66)
Shared-base decision	SEPARATE_ARTWORK_REQUIRED — CC-11.7B: split from the original unit202.generator.rotating-loop (2 states on 1 asset). The loop's face-on vs edge-on pose is a materially different silhouette/scene, not a labelling or overlay difference -- no deterministic overlay can rotate a loop's rendered 3D pose. Confirms brief §5's own invited scrutiny: reuse would have been chosen merely to reduce workload, not because it is pedagogically or technically honest.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/generator-rotating-loop-horizontal-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#49 **Simple rotating-loop AC generator — loop plane edge-on to poles — near-peak EMF** REQUIRED READY

`unit202.generator.rotating-loop.vertical` · family: Fleming's right-hand rule / AC generator effect · LO5 — lesson.emf.ac-generation-principles · role: PHENOMENON · class: HYBRID



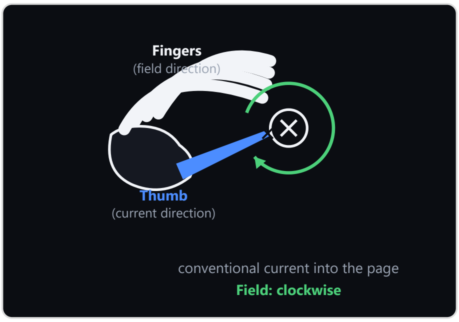
Show a single loop of wire, loop plane edge-on to the poles (loop seen from the side, near its thinnest silhouette), rotating on a central axis between N and S poles, establishing the physical basis of single-loop AC generation at Level 2 depth.

Reference	Wikimedia Commons — Diagram of single-phase generator with two poles.svg — https://commons.wikimedia.org/wiki/File:Diagram_of_single-phase_generator_with_two_poles.svg (CC0/public-domain reference material where recorded)
Immutable facts	N/S magnetic poles; loop/coil between the poles; loop plane edge-on to the poles (loop seen from the side, near its thinnest silhouette) -- this specific pose is the defining physical fact of this asset, never mixed with the sibling pose asset; central rotational axis; output/slip-ring concept at governed Level-2 abstraction; rotating conductor cuts magnetic flux
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Loop plane edge-on to poles — near-peak EMF [MULTI_STATE] (historical-66)
Shared-base decision	SEPARATE_ARTWORK_REQUIRED — CC-11.7B: split from the original unit202.generator.rotating-loop (2 states on 1 asset). The loop's face-on vs edge-on pose is a materially different silhouette/scene, not a labelling or overlay difference -- no deterministic overlay can rotate a loop's rendered 3D pose. Confirms brief §5's own invited scrutiny: reuse would have been chosen merely to reduce workload, not because it is pedagogically or technically honest.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/generator-rotating-loop-vertical-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

LO5 — lesson.magnetism.effects-of-current

#01 **Magnetic field around a current-carrying conductor** REQUIRED READY

`unit202.current-conductor.magnetic-field` · family: Right-hand grip rule / field around a conductor · LO5 — lesson.magnetism.effects-of-current · role: PHENOMENON · class: HYBRID



Show the actual electromagnetic phenomenon that the right-hand grip mnemonic predicts, independent of the mnemonic itself.

Reference	Wikimedia Commons — Right-hand grip rule.svg (field-line geometry only, not the hand) — https://commons.wikimedia.org/wiki/File:Right-hand_grip_rule.svg (Public-domain dedication)
Immutable facts	straight current-carrying conductor; magnetic field circulates around the conductor (closed concentric loops, not radiating outward); reversing the current reverses the direction of circulation; any depiction of field strength must not contradict a stronger field nearer the conductor; the concentric field-line pattern is rotationally symmetric -- do not bake in a specific current direction or circulation sense
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Current into page — field circulation revealed (teaching) [TEACHING] (historical-66) Current into page — field circulation withheld (assessment) [ASSESSMENT] (historical-66) Current out of page — field circulation revealed (teaching) [TEACHING] (historical-66) Current out of page — field circulation withheld (assessment) [ASSESSMENT] (historical-66)
Shared-base decision	SHARED_BASE_WITH_CONDITIONS — CC-11.7B: a straight conductor's concentric field-line pattern is rotationally symmetric -- reversing current only reverses which way the circulation arrows point, never the field-line geometry, conductor position, or composition. The base artwork itself is produced ONCE, direction-neutral, and the two current directions are distinguished entirely by a deterministic dot/cross + arrowhead OVERLAY added afterward -- the generated artwork is never mirrored, flipped, or redrawn to represent the opposite direction (CC-11.7C §2: mirroring hand-rule/directional artwork is prohibited outright, since it can silently invalidate a geometry-dependent mnemonic or phenomenon; this asset never relies on it). CORRECTION APPLIED (CC-11.7B): the pre-audit config incorrectly asked the art session to bake in 'current direction indicated' as part of the artwork (exactDeliverable) while also requiring 4 states covering both directions from one image -- an internal inconsistency. Fixed by moving both direction indicators to deterministicOverlayResponsibilities and removing direction-specific language from exactDeliverable/immutableFacts.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/current-conductor-magnetic-field-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#02 **Right-hand grip rule — teaching mnemonic** REQUIRED READY

`unit202.right-hand-grip.teaching` · family: Right-hand grip rule / field around a conductor · LO5 — lesson.magnetism.effects-of-current · role: MNEMONIC · class: HYBRID

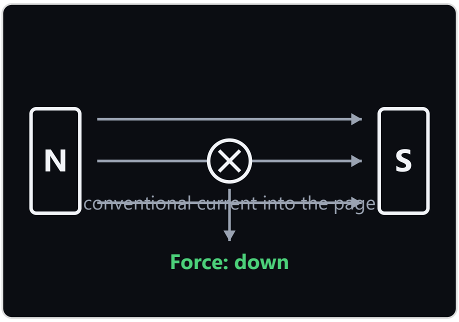
ARTWORK NOT YET PRODUCED

Teach that gripping a current-carrying conductor with the right hand, thumb along conventional current direction, gives the direction the magnetic field circulates as shown by the curled fingers.

Reference	Wikimedia Commons — Right-hand grip rule.svg — https://commons.wikimedia.org/wiki/File:Right-hand_grip_rule.svg (Public-domain dedication)
Immutable facts	RIGHT hand; thumb = conventional current direction; curled fingers = magnetic-field circulation direction; reversing current reverses magnetic-field rotation; straight conductor axis
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Right-hand grip mnemonic (teaching only) [TEACHING]
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B/CC-11.7C: single canonical state, single learner-facing object (one hand demonstrating one rule) -- no sharing/composite question applies. One base image remains sufficient because the mnemonic teaches the RULE itself (thumb = current, curled fingers = field), not a specific current direction. CC-11.7C correction: this is NOT achieved by mirroring or flipping the hand artwork for a reversed-current companion image -- that approach is explicitly prohibited (mirroring can silently change handedness and invalidate the mnemonic). Directional application for a specific current direction continues entirely through the sibling governed PHENOMENON/deterministic states (unit202.current-conductor.magnetic-field), never through a transformed copy of this mnemonic image.
Expected output file	`apps/mobile/src/assets/instructional/unit202/teaching/right-hand-grip-teaching-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#03 **Motor effect — conductor in magnetic field (N/S poles arranged horizontally)** REQUIRED BLOCKED

`unit202.motor.effect.horizontal-poles` · family: Fleming's left-hand rule / motor effect · LO5 — lesson.magnetism.effects-of-current · role: PHENOMENON · class: HYBRID



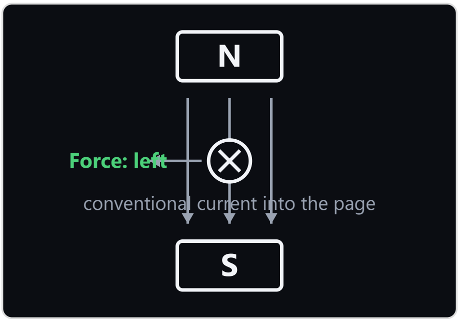
BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

Show a current-carrying conductor between magnetic poles (N/S poles arranged horizontally) experiencing a force perpendicular to both the field and the current (the motor effect), distinct from the Fleming's-left-hand mnemonic itself.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	N to S field direction; conductor positioned between the poles; poles arranged horizontally -- this specific orientation is the defining physical fact of this asset, never mixed with the sibling orientation asset; current explicitly into or out of the page -- the base artwork must not bake in either direction (deterministic overlay per state); resulting force perpendicular to both field and current; must remain visually distinct from the hand-rule mnemonic asset
Annotation policy	TEACHING_EXPLANATORY
Canonical states	N/S/horizontal poles, current into page — force revealed (teaching) [TEACHING] (historical-66) N/S/horizontal poles, current into page — force withheld (assessment) [ASSESSMENT] (historical-66) N/S/horizontal poles, current out of page — force revealed (teaching) [TEACHING] (historical-66) N/S/horizontal poles, current out of page — force withheld (assessment) [ASSESSMENT] (historical-66)
Shared-base decision	SHARED_BASE_WITH_CONDITIONS — CC-11.7B: split from the original unit202.motor.effect (8 states, all 4 pole/current combinations on one asset). Pole orientation is a genuine apparatus-layout change requiring separate artwork; current direction within one fixed orientation is a deterministic overlay concern, matching the right-hand-grip conductor asset's reasoning.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/motor-effect-horizontal-poles-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#48 **Motor effect — conductor in magnetic field (N/S poles arranged vertically)** REQUIRED BLOCKED

`unit202.motor.effect.vertical-poles` · family: Fleming's left-hand rule / motor effect · LO5 — lesson.magnetism.effects-of-current · role: PHENOMENON · class: HYBRID



BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

Show a current-carrying conductor between magnetic poles (N/S poles arranged vertically) experiencing a force perpendicular to both the field and the current (the motor effect), distinct from the Fleming's-left-hand mnemonic itself.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	N to S field direction; conductor positioned between the poles; poles arranged vertically -- this specific orientation is the defining physical fact of this asset, never mixed with the sibling orientation asset; current explicitly into or out of the page -- the base artwork must not bake in either direction (deterministic overlay per state); resulting force perpendicular to both field and current; must remain visually distinct from the hand-rule mnemonic asset
Annotation policy	TEACHING_EXPLANATORY
Canonical states	N/S/vertical poles, current into page — force revealed (teaching) [TEACHING] (historical-66) N/S/vertical poles, current into page — force withheld (assessment) [ASSESSMENT] (historical-66) N/S/vertical poles, current out of page — force revealed (teaching) [TEACHING] (historical-66) N/S/vertical poles, current out of page — force withheld (assessment) [ASSESSMENT] (historical-66)
Shared-base decision	SHARED_BASE_WITH_CONDITIONS — CC-11.7B: split from the original unit202.motor.effect (8 states, all 4 pole/current combinations on one asset). Pole orientation is a genuine apparatus-layout change requiring separate artwork; current direction within one fixed orientation is a deterministic overlay concern, matching the right-hand-grip conductor asset's reasoning.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/motor-effect-vertical-poles-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#04 **Fleming left-hand rule — motor teaching mnemonic** REQUIRED READY

`unit202.fleming-left-hand.teaching` · family: Fleming's left-hand rule / motor effect · LO5 — lesson.magnetism.effects-of-current · role: MNEMONIC · class: HYBRID

ARTWORK NOT YET PRODUCED

Teach the motor-effect hand mnemonic: thuMb = Motion/force, First finger = Field, seCond finger = Current, each mutually perpendicular on the left hand.

Reference	Wikimedia Commons — Fleming's Left Hand Rule.png — https://commons.wikimedia.org/wiki/File:Fleming%27s_Left_Hand_Rule.png (Openly licensed -- treat primarily as geometry/reference, not artwork to imitate)
Immutable facts	LEFT hand; thumb = force/motion; first (index) finger = magnetic FIELD; second (middle) finger = conventional CURRENT; mutually perpendicular relationship between all three
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Fleming's left-hand mnemonic (teaching only) [TEACHING]
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: single canonical state, single learner-facing object -- no sharing/composite question applies. The mnemonic teaches the Force/Field/Current correspondence itself, not a specific pole/current combination, so it does not need a companion image per pole-orientation the way the sibling PHENOMENON assets do. CC-11.7C: this is never achieved by mirroring/flipping the hand artwork -- see prohibitedChanges.
Expected output file	`apps/mobile/src/assets/instructional/unit202/teaching/fleming-left-hand-teaching-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#06 **Fleming right-hand rule — generator teaching mnemonic** REQUIRED READY

`unit202.fleming-right-hand.teaching` · family: Fleming's right-hand rule / AC generator effect · LO5 — lesson.magnetism.effects-of-current · role: MNEMONIC · class: HYBRID

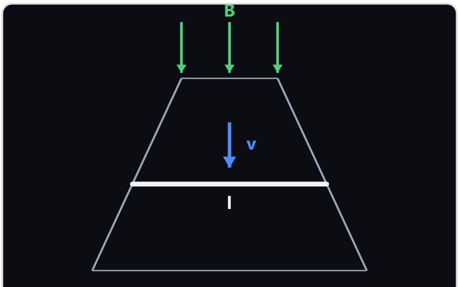
ARTWORK NOT YET PRODUCED

Teach the generator-effect hand mnemonic: thuMb = Motion of the conductor, First finger = Field, seCond finger = induced Current/EMF, each mutually perpendicular on the right hand.

Reference	Fleming's original 1902 right-hand-rule figure (Wikimedia Commons) — https://commons.wikimedia.org/wiki/File:Fleming%27s_right_hand_rule.png (Public-domain historical work)
Immutable facts	RIGHT hand; thumb = conductor MOTION; first (index) finger = magnetic FIELD; second (middle) finger = induced current/EMF; three directions mutually perpendicular
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Fleming's right-hand mnemonic (teaching only) [TEACHING]
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: single canonical state, single learner-facing object -- no sharing/composite question applies. The mnemonic teaches the Motion/Field/induced-Current correspondence itself, not a specific loop pose, so it does not need a companion image per rotation phase the way the sibling PHENOMENON assets do. CC-11.7C: this is never achieved by mirroring/flipping the hand artwork -- see prohibitedChanges.
Expected output file	`apps/mobile/src/assets/instructional/unit202/teaching/fleming-right-hand-teaching-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#22 **Motional EMF geometry** REQUIRED BLOCKED

`unit202.emf.motional` · family: Motional EMF geometry · LO5 — lesson.magnetism.effects-of-current · role: PHENOMENON · class: HYBRID



B, l and v are mutually perpendicular

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

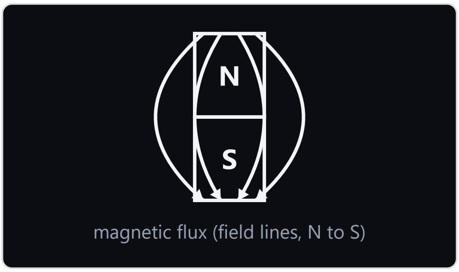
Show that conductor length, its velocity and the magnetic field are mutually perpendicular -- the geometric fact behind e = Blv.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	B, l and v mutually perpendicular for the governed e = Blv case; rod across rails; velocity along the rails; field perpendicular to the rail plane
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Motional EMF — B, l, v mutually perpendicular [MULTI_STATE] (historical-66)
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/emf-motional-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO5 — lesson.magnetism.fundamentals

#12 **Bar magnet and magnetic field** REQUIRED READY

`unit202.magnet.field` · family: Magnetism — field and pole interaction · LO5 — lesson.magnetism.fundamentals · role: PHENOMENON · class: HYBRID

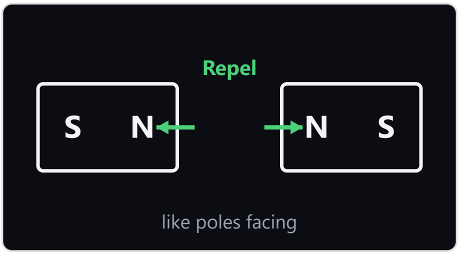


Show a bar magnet with its external magnetic field pattern, N to S, as the basis for flux/flux-density teaching.

Reference	Wikimedia Commons — DipolMagnet.svg — https://commons.wikimedia.org/wiki/File:DipolMagnet.svg (Public-domain dedication)
Immutable facts	meaningful field-line geometry; external field direction runs N to S; field-line density used meaningfully where density is taught
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Bar magnet field pattern (no density comparison) [MULTI_STATE] (historical-66) Bar magnet field with flux-density comparison (same flux, different cross-section) [MULTI_STATE] (historical-66)
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: the density-comparison state adds a deterministic comparison callout to the same bar magnet + field-line composition -- the magnet's position and field-line geometry are unchanged. The base artwork already documents this as an overlay concern (exactDeliverable: 'ready to receive a deterministic N/S-labelled field-line overlay').
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/magnet-field-base-v{N}.(png|webp|jpg)`
Studio prompt	Available in Studio (COPY PROMPT / VIEW PROMPT)

#13 **Magnetic pole repulsion — like poles facing (repel)** REQUIRED BLOCKED

`unit202.magnet.poles.like` · family: Magnetism — field and pole interaction · LO5 — lesson.magnetism.fundamentals · role: COMPARISON · class: HYBRID



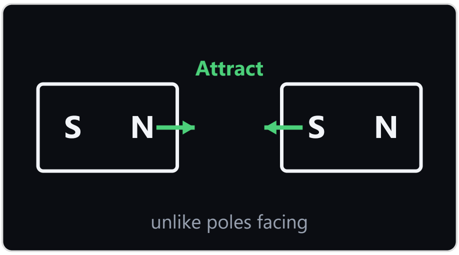
BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

Show like poles facing (repel) from the pole labels on two bar magnets.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	like poles repel; magnets must be composed with a spacing appropriate to repulsion (visible gap, compressed/deflected field lines) -- this specific spacing is the defining physical fact of this asset, never mixed with the sibling asset; field behaviour between the two magnets must remain physically meaningful
Annotation policy	TEACHING_EXPLANATORY
Canonical states	like poles facing (repel) — force revealed (teaching) [TEACHING] (historical-66) like poles facing (repel) — force withheld (assessment) [ASSESSMENT] (historical-66)
Shared-base decision	SEPARATE_ARTWORK_REQUIRED — CC-11.7B: split from the original unit202.magnet.poles (4 states on 1 asset whose own exactDeliverable already literally asked for 'Two premium illustrations' in one prompt). Attracting vs repelling poles are conventionally composed with genuinely different magnet spacing/field-line behaviour, not merely a different arrow direction on identical geometry.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/magnet-poles-like-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#50 **Magnetic pole attraction — unlike poles facing (attract)** REQUIRED BLOCKED

`unit202.magnet.poles.unlike` · family: Magnetism — field and pole interaction · LO5 — lesson.magnetism.fundamentals · role: COMPARISON · class: HYBRID



BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

Show unlike poles facing (attract) from the pole labels on two bar magnets.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	unlike poles attract; magnets must be composed with a spacing appropriate to attraction (close together, merging field lines) -- this specific spacing is the defining physical fact of this asset, never mixed with the sibling asset; field behaviour between the two magnets must remain physically meaningful
Annotation policy	TEACHING_EXPLANATORY
Canonical states	unlike poles facing (attract) — force revealed (teaching) [TEACHING] (historical-66) unlike poles facing (attract) — force withheld (assessment) [ASSESSMENT] (historical-66)
Shared-base decision	SEPARATE_ARTWORK_REQUIRED — CC-11.7B: split from the original unit202.magnet.poles (4 states on 1 asset whose own exactDeliverable already literally asked for 'Two premium illustrations' in one prompt). Attracting vs repelling poles are conventionally composed with genuinely different magnet spacing/field-line behaviour, not merely a different arrow direction on identical geometry.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/magnet-poles-unlike-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#43 **Permanent magnet vs electromagnet comparison** USEFUL BLOCKED

`unit202.magnet.permanent-vs-electromagnet` · family: Magnetism — field and pole interaction · LO5 — lesson.magnetism.fundamentals · role: COMPARISON · class: HYBRID

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

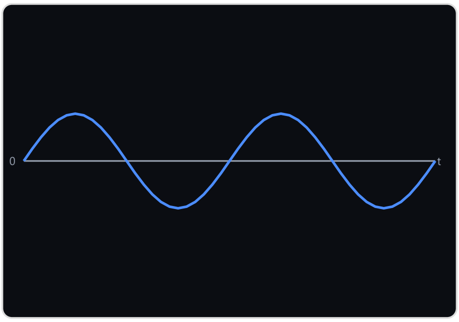
CC-11.7 audit finding: a genuine physical-topology comparison (coiled wire around a core vs a solid bar magnet) the learner can use to distinguish the two magnet types by appearance -- USEFUL enrichment, Level 2 depth keeps it below REQUIRED.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	permanent magnet depicted as a solid bar/horseshoe magnet, no coil or power source; electromagnet depicted as a coil of wire around a core with a visible power source/current path; one side-by-side comparison image, not two separate images
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Permanent magnet vs electromagnet (blocked) [TEACHING]
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B composite-image audit (§8): the intended learner-facing deliverable is deliberately ONE side-by-side comparison visual (permanent magnet | electromagnet) -- the comparison itself is the teaching objective, matching the brief's own explicit 'ACCEPTABLE' example. Confirmed as one ProductionAsset, not split.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/magnet-permanent-vs-electromagnet-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO5 — lesson.waveforms.ac-dc-and-sine-wave-quantities

#21 **AC sine waveform** REQUIRED NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.waveform.sine` · family: AC sine waveform · LO5 — lesson.waveforms.ac-dc-and-sine-wave-quantities · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL



A visually polished sine-waveform reference for style/contrast QA against the existing deterministic renderer -- waveform correctness is already governed.

Reference	Wikimedia Commons — Sine wave 2.svg — https://commons.wikimedia.org/wiki/File:Sine_wave_2.svg (Public-domain dedication)
Immutable facts	mathematically correct sine curve; zero axis shown; peak shown; peak-to-peak shown; period shown; RMS is approximately 0.707 x peak where taught
Annotation policy	NONE
Canonical states	Sine waveform — bare cycle (zero-axis reference only) (2 cycles) [MULTI_STATE] (historical-66) Sine waveform — peak revealed (2 cycles) [MULTI_STATE] (historical-66) Sine waveform — peak + RMS revealed (2 cycles) [MULTI_STATE] (historical-66) Sine waveform — peak + RMS + period revealed (2 cycles) [MULTI_STATE] (historical-66) Sine waveform — single cycle, fully annotated (1 cycle) [MULTI_STATE] (historical-66) Sine waveform — three cycles, fully annotated (3 cycles) [MULTI_STATE] (historical-66)
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: DETERMINISTIC_TECHNICAL -- peak/RMS/period-marker/cycle-count states are deterministic vector plots, no generated artwork involved at all.
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO6 — lesson.electrical.electronic-components-passive

#30 **Diode forward bias — conducting** REQUIRED BLOCKED

`unit202.diode.bias-direction.forward` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-passive · role: PHENOMENON · class: HYBRID

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

CC-11.7 audit finding (new, beyond the original 66): show current flowing easily in forward bias, distinct from the static IEC symbol -- directly targets EL-COMPONENT-DIODE-001 and the named misconception MIS-EL-DIODE-DIRECTION-CONFUSION-001 (confusing which direction a diode conducts), which the existing `electronics.component_symbol_card` blueprint cannot represent since it renders only the static symbol, never current flow.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	diode conducts easily in forward bias (anode more positive than cathode); must remain visually distinct from the plain IEC diode symbol asset; must depict ONLY the forward-bias state -- the sibling ProductionAsset covers the other
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Forward bias — conducting [TEACHING]
Shared-base decision	SEPARATE_ARTWORK_REQUIRED — CC-11.7B: split from the original unit202.diode.bias-direction (2 states on 1 asset whose own exactDeliverable already literally asked for 'Two premium illustrations' in one prompt). Conducting vs blocked current flow is a genuine presence/absence scene difference (visible current-flow indication vs a visibly blocked path), not a label swap on identical geometry.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/diode-bias-direction-forward-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#53 **Diode reverse bias — blocked** REQUIRED BLOCKED

`unit202.diode.bias-direction.reverse` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-passive · role: PHENOMENON · class: HYBRID

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

CC-11.7 audit finding (new, beyond the original 66): show current blocked in reverse bias, distinct from the static IEC symbol -- directly targets EL-COMPONENT-DIODE-001 and the named misconception MIS-EL-DIODE-DIRECTION-CONFUSION-001 (confusing which direction a diode conducts), which the existing `electronics.component_symbol_card` blueprint cannot represent since it renders only the static symbol, never current flow.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	diode blocks current in reverse bias (cathode more positive than anode); must remain visually distinct from the plain IEC diode symbol asset; must depict ONLY the reverse-bias state -- the sibling ProductionAsset covers the other
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Reverse bias — blocked [TEACHING]
Shared-base decision	SEPARATE_ARTWORK_REQUIRED — CC-11.7B: split from the original unit202.diode.bias-direction (2 states on 1 asset whose own exactDeliverable already literally asked for 'Two premium illustrations' in one prompt). Conducting vs blocked current flow is a genuine presence/absence scene difference (visible current-flow indication vs a visibly blocked path), not a label swap on identical geometry.
Expected output file	`apps/mobile/src/assets/instructional/unit202/hybrid/diode-bias-direction-reverse-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#32 **Capacitor RC charge/discharge transient curve** REQUIRED NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.capacitor.transient` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-passive · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL

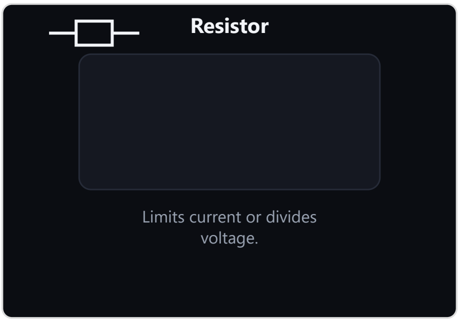
CC-11.7 audit finding (new, beyond the original 66): the exponential RC charge/discharge curve directly targeted by EL-COMPONENT-CAPACITOR-TRANSIENT-001 and question blueprint `electronics.recognise_capacitor_behaviour` ('gradual_exponential_change' vs 'instant_step_change'), which no existing blueprint depicts -- an exponential curve is a genuinely different shape from the plain sine wave.

Reference	Standard RC charge/discharge exponential-curve reference -- to be selected when this asset is commissioned (to be recorded when selected)
Immutable facts	charge/discharge follows a genuine exponential curve, never a straight-line ramp or instant step; never a sine wave -- this is a transient response, not a periodic waveform
Annotation policy	NONE
Canonical states	Charge curve (exponential rise) [MULTI_STATE] Discharge curve (exponential decay) [MULTI_STATE]
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: DETERMINISTIC_TECHNICAL -- charge/discharge exponential curves are deterministic vector plots, no generated artwork involved at all.
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO6 — lesson.electrical.electronic-components-passive / -switching-control

#23 **Electronic component symbol system** REQUIRED NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.components.symbols` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-passive / -switching-control · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL



The governed UK/IEC schematic-symbol system for component recognition -- symbol geometry must never be AI-generated.

Reference	IEC 60617 graphical-symbol system / current UK technical-drawing convention (standards reference -- verify against the current governed BS EN 60617 / IEC 60617 convention)
Immutable facts	every symbol must match the governed BS EN 60617 / IEC 60617 convention exactly
Annotation policy	NONE
Canonical states	UK/IEC symbol — resistor [MULTI_STATE] (historical-66) UK/IEC symbol — capacitor [MULTI_STATE] (historical-66) UK/IEC symbol — diode [MULTI_STATE] (historical-66) UK/IEC symbol — zener diode [MULTI_STATE] (historical-66) UK/IEC symbol — led [MULTI_STATE] (historical-66) UK/IEC symbol — photodiode [MULTI_STATE] (historical-66) UK/IEC symbol — thermistor [MULTI_STATE] (historical-66) UK/IEC symbol — diac [MULTI_STATE] (historical-66) UK/IEC symbol — triac [MULTI_STATE] (historical-66) UK/IEC symbol — transistor [MULTI_STATE] (historical-66) UK/IEC symbol — thyristor scr [MULTI_STATE] (historical-66) UK/IEC symbol — rectifier [MULTI_STATE] (historical-66) UK/IEC symbol — inverter [MULTI_STATE] (historical-66)
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: DETERMINISTIC_TECHNICAL -- governed UK/IEC symbol geometry, produced by ComponentSymbols.tsx, never an art session. No generated-artwork sharing question applies.
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#24 **Physical electronic component — resistor** REQUIRED BLOCKED

`unit202.components.physical.resistor` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-passive / -switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the resistor, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a resistor
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — resistor [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-resistor-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#25 **Physical electronic component — capacitor**

REQUIRED BLOCKED

`unit202.components.physical.capacitor` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-passive / -switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the capacitor, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a capacitor
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — capacitor [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-capacitor-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#26 **Physical electronic component — diode**

REQUIRED BLOCKED

`unit202.components.physical.diode` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-passive / -switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the diode, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a diode
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — diode [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-diode-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#27 **Physical electronic component — led**

REQUIRED BLOCKED

`unit202.components.physical.led` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-passive / -switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the led, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a led
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — led [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-led-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#28 **Physical electronic component — thermistor** REQUIRED BLOCKED

`unit202.components.physical.thermistor` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-passive / -switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the thermistor, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a thermistor
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — thermistor [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-thermistor-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#29 **Physical electronic component — transistor** REQUIRED BLOCKED

`unit202.components.physical.transistor` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-passive / -switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the transistor, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a transistor
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — transistor [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-transistor-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

LO6 — lesson.electrical.electronic-components-switching-control

#31 Rectifier/inverter output waveform shapes

REQUIRED | NO CHATGPT ART JOB / VECTOR AUTHORITATIVE

`unit202.rectification.waveforms` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-switching-control · role: TECHNICAL_DIAGRAM · class: DETERMINISTIC_TECHNICAL

CC-11.7 audit finding (new, beyond the original 66): the three distinct output-waveform SHAPES (half-wave rectified, full-wave rectified, inverter-synthesised AC) directly targeted by question blueprint `electronics.recognise_rectifier_type`, which the existing `graph.waveform_sine` blueprint cannot represent (it renders only a plain sine wave) and the existing `electronics.component_symbol_card`'s rectifier/inverter entries cannot represent either (they render only the functional-block symbol, never the resulting waveform shape).

Reference	Standard half-wave/full-wave rectification and inverter output waveform references -- to be selected when this asset is commissioned (to be recorded when selected)
Immutable facts	half-wave: blocks one half-cycle entirely, passes the other unchanged in shape; full-wave: converts both half-cycles to the same polarity (pulsating DC, never a flat line); inverter: DC input synthesised into an AC-shaped output via controlled switching; never a smooth sine wave for any of the three -- that is the plain graph.waveform_sine blueprint's own separate, correct depiction
Annotation policy	NONE
Canonical states	Half-wave rectified output [MULTI_STATE] Full-wave rectified output [MULTI_STATE] Inverter-synthesised AC output [MULTI_STATE]
Shared-base decision	SAFE_SHARED_BASE — CC-11.7B: DETERMINISTIC_TECHNICAL -- half-wave/full-wave/inverter waveform shapes are deterministic vector plots, no generated artwork involved at all.
Expected output file	`SVG / React Native SVG (deterministic vector geometry) -- VECTOR AUTHORITATIVE, no raster production file`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#33 Physical electronic component — zener diode

USEFUL | BLOCKED

`unit202.components.physical.zener-diode` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the zener diode, paired with its existing deterministic UK/IEC symbol card -- more specialist than the six REQUIRED components, genuinely useful but not REQUIRED for Unit 202 visual completeness (CC-11.7 audit finding).

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a zener diode
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — zener diode (blocked) [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-zener-diode-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#34 **Physical electronic component — photodiode** USEFUL BLOCKED

`unit202.components.physical.photodiode` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the photodiode, paired with its existing deterministic UK/IEC symbol card -- more specialist than the six REQUIRED components, genuinely useful but not REQUIRED for Unit 202 visual completeness (CC-11.7 audit finding).

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a photodiode
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — photodiode (blocked) [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-photodiode-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#35 **Physical electronic component — DIAC** USEFUL BLOCKED

`unit202.components.physical.diac` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the DIAC, paired with its existing deterministic UK/IEC symbol card -- more specialist than the six REQUIRED components, genuinely useful but not REQUIRED for Unit 202 visual completeness (CC-11.7 audit finding).

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a DIAC
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — DIAC (blocked) [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-diac-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#36 Physical electronic component — TRIAC

USEFUL BLOCKED

`unit202.components.physical.triac` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the TRIAC, paired with its existing deterministic UK/IEC symbol card -- more specialist than the six REQUIRED components, genuinely useful but not REQUIRED for Unit 202 visual completeness (CC-11.7 audit finding).

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a TRIAC
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — TRIAC (blocked) [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-triac-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

#37 Physical electronic component — thyristor/SCR

USEFUL BLOCKED

`unit202.components.physical.thyristor-scr` · family: Electronic components — recognition · LO6 — lesson.electrical.electronic-components-switching-control · role: PHYSICAL_RECOGNITION · class: PREMIUM_CONCEPTUAL

ARTWORK NOT YET PRODUCED

BLOCKED — REFERENCE REQUIRED BEFORE PRODUCTION.

A physical-appearance companion image for the thyristor/SCR, paired with its existing deterministic UK/IEC symbol card -- more specialist than the six REQUIRED components, genuinely useful but not REQUIRED for Unit 202 visual completeness (CC-11.7 audit finding).

Reference	PRIMARY REFERENCE STILL TO BE APPROVED (unknown -- not yet sourced)
Immutable facts	package form must be a real, representative physical form for a thyristor/SCR
Annotation policy	TEACHING_EXPLANATORY
Canonical states	Physical appearance — thyristor/SCR (blocked) [TEACHING]
Shared-base decision	single state / not applicable
Expected output file	`apps/mobile/src/assets/instructional/unit202/physical-components/components-physical-thyristor-scr-base-v{N}.(png|webp|jpg)`
Studio prompt	Not offered (blocked, scope-pending, or deterministic-only)

Multi-state sharing review

Every ProductionAsset with more than one CanonicalState, audited for whether the proposed image sharing is genuinely safe. 30 assets audited: 18 kept, 3 kept with conditions, 9 split.

Asset	States	Decision	Rationale	Action
`unit202.current-conductor.magnetic-field` Magnetic field around a current-carrying conductor	4	**SHARED_BASE_WITH_CONDITIONS**	CC-11.7B: a straight conductor's concentric field-line pattern is rotationally symmetric -- reversing current only reverses which way the circulation arrows point, never the field-line geometry, conductor position, or composition. The base artwork itself is produced ONCE, direction-neutral, and the two current directions are distinguished entirely by a deterministic dot/cross + arrowhead OVERLAY added afterward -- the generated artwork is never mirrored, flipped, or redrawn to represent the opposite direction (CC-11.7C §2: mirroring hand-rule/directional artwork is prohibited outright, since it can silently invalidate a geometry-dependent mnemonic or phenomenon; this asset never relies on it). CORRECTION APPLIED (CC-11.7B): the pre-audit config incorrectly asked the art session to bake in 'current direction indicated' as part of the artwork (exactDeliverable) while also requiring 4 states covering both directions from one image -- an internal inconsistency. Fixed by moving both direction indicators to deterministicOverlayResponsibilities and removing direction-specific language from exactDeliverable/immutableFacts.	KEEP_WITH_CONDITIONS
`unit202.right-hand-grip.teaching` Right-hand grip rule — teaching mnemonic	1	**SAFE_SHARED_BASE**	CC-11.7B/CC-11.7C: single canonical state, single learner-facing object (one hand demonstrating one rule) -- no sharing/composite question applies. One base image remains sufficient because the mnemonic teaches the RULE itself (thumb = current, curled fingers = field), not a specific current direction. CC-11.7C correction: this is NOT achieved by mirroring or flipping the hand artwork for a reversed-current companion image -- that approach is explicitly prohibited (mirroring can silently change handedness and invalidate the mnemonic). Directional application for a specific current direction continues entirely through the sibling governed PHENOMENON/deterministic states (unit202.current-conductor.magnetic-field), never through a transformed copy of this mnemonic image.	KEEP

Asset	States	Decision	Rationale	Action
`unit202.motor.effect.horizontal-poles` Motor effect — conductor in magnetic field (N/S poles arranged horizontally)	4	**SHARED_BASE_WITH_CONDITIONS**	CC-11.7B: split from the original unit202.motor.effect (8 states, all 4 pole/current combinations on one asset). Pole orientation is a genuine apparatus-layout change requiring separate artwork; current direction within one fixed orientation is a deterministic overlay concern, matching the right-hand-grip conductor asset's reasoning.	KEEP_WITH_CONDITIONS
`unit202.motor.effect.vertical-poles` Motor effect — conductor in magnetic field (N/S poles arranged vertically)	4	**SHARED_BASE_WITH_CONDITIONS**	CC-11.7B: split from the original unit202.motor.effect (8 states, all 4 pole/current combinations on one asset). Pole orientation is a genuine apparatus-layout change requiring separate artwork; current direction within one fixed orientation is a deterministic overlay concern, matching the right-hand-grip conductor asset's reasoning.	KEEP_WITH_CONDITIONS
`unit202.fleming-left-hand.teaching` Fleming left-hand rule — motor teaching mnemonic	1	**SAFE_SHARED_BASE**	CC-11.7B: single canonical state, single learner-facing object -- no sharing/composite question applies. The mnemonic teaches the Force/Field/Current correspondence itself, not a specific pole/current combination, so it does not need a companion image per pole-orientation the way the sibling PHENOMENON assets do. CC-11.7C: this is never achieved by mirroring/flipping the hand artwork -- see prohibitedChanges.	KEEP
`unit202.generator.rotating-loop.horizontal` Simple rotating-loop AC generator — loop plane facing poles — near-zero EMF	1	**SEPARATE_ARTWORK_REQUIRED**	CC-11.7B: split from the original unit202.generator.rotating-loop (2 states on 1 asset). The loop's face-on vs edge-on pose is a materially different silhouette/scene, not a labelling or overlay difference -- no deterministic overlay can rotate a loop's rendered 3D pose. Confirms brief §5's own invited scrutiny: reuse would have been chosen merely to reduce workload, not because it is pedagogically or technically honest.	SPLIT
`unit202.generator.rotating-loop.vertical` Simple rotating-loop AC generator — loop plane edge-on to poles — near-peak EMF	1	**SEPARATE_ARTWORK_REQUIRED**	CC-11.7B: split from the original unit202.generator.rotating-loop (2 states on 1 asset). The loop's face-on vs edge-on pose is a materially different silhouette/scene, not a labelling or overlay difference -- no deterministic overlay can rotate a loop's rendered 3D pose. Confirms brief §5's own invited scrutiny: reuse would have been chosen merely to reduce workload, not because it is pedagogically or technically honest.	SPLIT

Asset	States	Decision	Rationale	Action
`unit202.fleming-right-hand.teaching` Fleming right-hand rule — generator teaching mnemonic	1	**SAFE_SHARED_BASE**	CC-11.7B: single canonical state, single learner-facing object -- no sharing/composite question applies. The mnemonic teaches the Motion/Field/induced-Current correspondence itself, not a specific loop pose, so it does not need a companion image per rotation phase the way the sibling PHENOMENON assets do. CC-11.7C: this is never achieved by mirroring/flipping the hand artwork -- see prohibitedChanges.	KEEP
`unit202.levers.class-1` Lever — Class I	2	**SAFE_SHARED_BASE**	CC-11.7B: 'recognition' and 'moment-balance' states depict the identical physical lever rig (same fulcrum/effort/load positions) -- the only difference is whether deterministic effort-arm/load-arm distance annotations are added on top. No pose, configuration or scene change.	KEEP
`unit202.levers.class-2` Lever — Class II	2	**SAFE_SHARED_BASE**	CC-11.7B: 'recognition' and 'moment-balance' states depict the identical physical lever rig (same fulcrum/effort/load positions) -- the only difference is whether deterministic effort-arm/load-arm distance annotations are added on top. No pose, configuration or scene change.	KEEP
`unit202.levers.class-3` Lever — Class III	2	**SAFE_SHARED_BASE**	CC-11.7B: 'recognition' and 'moment-balance' states depict the identical physical lever rig (same fulcrum/effort/load positions) -- the only difference is whether deterministic effort-arm/load-arm distance annotations are added on top. No pose, configuration or scene change.	KEEP
`unit202.magnet.field` Bar magnet and magnetic field	2	**SAFE_SHARED_BASE**	CC-11.7B: the density-comparison state adds a deterministic comparison callout to the same bar magnet + field-line composition -- the magnet's position and field-line geometry are unchanged. The base artwork already documents this as an overlay concern (exactDeliverable: 'ready to receive a deterministic N/S-labelled field-line overlay').	KEEP
`unit202.magnet.poles.like` Magnetic pole repulsion — like poles facing (repel)	2	**SEPARATE_ARTWORK_REQUIRED**	CC-11.7B: split from the original unit202.magnet.poles (4 states on 1 asset whose own exactDeliverable already literally asked for 'Two premium illustrations' in one prompt). Attracting vs repelling poles are conventionally composed with genuinely different magnet spacing/field-line behaviour, not merely a different arrow direction on identical geometry.	SPLIT

Asset	States	Decision	Rationale	Action
`unit202.magnet.poles.unlike` Magnetic pole attraction — unlike poles facing (attract)	2	**SEPARATE_ARTWORK_REQUIRED**	CC-11.7B: split from the original unit202.magnet.poles (4 states on 1 asset whose own exactDeliverable already literally asked for 'Two premium illustrations' in one prompt). Attracting vs repelling poles are conventionally composed with genuinely different magnet spacing/field-line behaviour, not merely a different arrow direction on identical geometry.	SPLIT
`unit202.magnet.permanent-vs-electromagnet` Permanent magnet vs electromagnet comparison	1	**SAFE_SHARED_BASE**	CC-11.7B composite-image audit (§8): the intended learner-facing deliverable is deliberately ONE side-by-side comparison visual (permanent magnet | electromagnet) -- the comparison itself is the teaching objective, matching the brief's own explicit 'ACCEPTABLE' example. Confirmed as one ProductionAsset, not split.	KEEP
`unit202.circuit.series` Series circuit	3	**SAFE_SHARED_BASE**	CC-11.7B: DETERMINISTIC_TECHNICAL -- component-count states are deterministic vector geometry, no generated artwork involved at all.	KEEP
`unit202.circuit.parallel` Parallel circuit	3	**SAFE_SHARED_BASE**	CC-11.7B: DETERMINISTIC_TECHNICAL -- branch-count states are deterministic vector geometry, no generated artwork involved at all.	KEEP
`unit202.circuit.mixed` Mixed series/parallel circuit	2	**SAFE_SHARED_BASE**	CC-11.7B: DETERMINISTIC_TECHNICAL -- arrangement states are deterministic vector topology, no generated artwork involved at all.	KEEP
`unit202.instrument.connections` Ammeter / voltmeter / ohmmeter connections	5	**SAFE_SHARED_BASE**	CC-11.7B: DETERMINISTIC_TECHNICAL -- instrument/connection-style states are deterministic vector topology, no generated artwork involved at all.	KEEP
`unit202.gears.driven-larger` Driver/driven gears — driven gear larger than the driver	1	**SEPARATE_ARTWORK_REQUIRED**	CC-11.7B: split from the original unit202.gears (3 states on 1 asset). Driven-gear size relative to the driver is a genuine physical size change to a rendered object, not a labelling/overlay difference -- no deterministic overlay mechanism can resize a gear in generated artwork (only rotation-direction is declared as an overlay responsibility).	SPLIT
`unit202.gears.driven-smaller` Driver/driven gears — driven gear smaller than the driver	1	**SEPARATE_ARTWORK_REQUIRED**	CC-11.7B: split from the original unit202.gears (3 states on 1 asset). Driven-gear size relative to the driver is a genuine physical size change to a rendered object, not a labelling/overlay difference -- no deterministic overlay mechanism can resize a gear in generated artwork (only rotation-direction is declared as an overlay responsibility).	SPLIT

Asset	States	Decision	Rationale	Action
`unit202.gears.equal` Driver/driven gears — driver and driven gears equal size	1	**SEPARATE_ARTWORK_REQUIRED**	CC-11.7B: split from the original unit202.gears (3 states on 1 asset). Driven-gear size relative to the driver is a genuine physical size change to a rendered object, not a labelling/overlay difference -- no deterministic overlay mechanism can resize a gear in generated artwork (only rotation-direction is declared as an overlay responsibility).	SPLIT
`unit202.gears.rotation-direction` Gear rotation-direction reversal / idler gear	2	**SAFE_SHARED_BASE**	CC-11.7B: purely deterministic annotation (rotation-direction arrows, idler presence) on whichever sibling driver/driven gear base artwork applies -- no generated artwork involved at all, unaffected by the CC-11.7B size-ratio split.	KEEP
`unit202.waveform.sine` AC sine waveform	6	**SAFE_SHARED_BASE**	CC-11.7B: DETERMINISTIC_TECHNICAL -- peak/RMS/period-marker/cycle-count states are deterministic vector plots, no generated artwork involved at all.	KEEP
`unit202.components.symbols` Electronic component symbol system	13	**SAFE_SHARED_BASE**	CC-11.7B: DETERMINISTIC_TECHNICAL -- governed UK/IEC symbol geometry, produced by ComponentSymbols.tsx, never an art session. No generated-artwork sharing question applies.	KEEP
`unit202.diode.bias-direction.forward` Diode forward bias — conducting	1	**SEPARATE_ARTWORK_REQUIRED**	CC-11.7B: split from the original unit202.diode.bias-direction (2 states on 1 asset whose own exactDeliverable already literally asked for 'Two premium illustrations' in one prompt). Conducting vs blocked current flow is a genuine presence/absence scene difference (visible current-flow indication vs a visibly blocked path), not a label swap on identical geometry.	SPLIT
`unit202.diode.bias-direction.reverse` Diode reverse bias — blocked	1	**SEPARATE_ARTWORK_REQUIRED**	CC-11.7B: split from the original unit202.diode.bias-direction (2 states on 1 asset whose own exactDeliverable already literally asked for 'Two premium illustrations' in one prompt). Conducting vs blocked current flow is a genuine presence/absence scene difference (visible current-flow indication vs a visibly blocked path), not a label swap on identical geometry.	SPLIT
`unit202.rectification.waveforms` Rectifier/inverter output waveform shapes	3	**SAFE_SHARED_BASE**	CC-11.7B: DETERMINISTIC_TECHNICAL -- half-wave/full-wave/inverter waveform shapes are deterministic vector plots, no generated artwork involved at all.	KEEP

Asset	States	Decision	Rationale	Action
`unit202.capacitor.transient` Capacitor RC charge/discharge transient curve	2	**SAFE_SHARED_BASE**	CC-11.7B: DETERMINISTIC_TECHNICAL -- charge/discharge exponential curves are deterministic vector plots, no generated artwork involved at all.	KEEP
`unit202.protective-devices` Fuse vs circuit breaker comparison	1	**SAFE_SHARED_BASE**	CC-11.7B §5/§8 explicit pedagogical decision: the comparison itself (fuse must be replaced; breaker can be reset) is the teaching objective cap.fault.compare_fuse_breaker asks for, matching the brief's own 'ACCEPTABLE: Fuse vs circuit breaker comparison where comparison itself is the teaching objective' example. Confirmed as one intentional composite ProductionAsset, not two independent physical-recognition assets. ROLE corrected from the pre-audit PHYSICAL_RECOGNITION (a mismatch for a two-object comparison) to COMPARISON.	KEEP

Actual image-generation job list — the real production workload

42 separate ChatGPT image-generation jobs (34 REQUIRED, 8 USEFUL). This is the Product Owner's real manual workload -- not the family count, not the state count, not the asset count.

#	Asset	Need	Status	Reference	States supported
ART JOB 01	`unit202.current-conductor.magnetic-field` Magnetic field around a current-carrying conductor	REQUIRED	READY	Wikimedia Commons — Right-hand grip rule.svg (field-line geometry only, not the hand)	Current into page — field circulation revealed (teaching) Current into page — field circulation withheld (assessment) Current out of page — field circulation revealed (teaching) Current out of page — field circulation withheld (assessment)
ART JOB 02	`unit202.right-hand-grip.teaching` Right-hand grip rule — teaching mnemonic	REQUIRED	READY	Wikimedia Commons — Right-hand grip rule.svg	Right-hand grip mnemonic (teaching only)
ART JOB 03	`unit202.motor.effect.horizontal-poles` Motor effect — conductor in magnetic field (N/S poles arranged horizontally)	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	N/S/horizontal poles, current into page — force revealed (teaching) N/S/horizontal poles, current into page — force withheld (assessment) N/S/horizontal poles, current out of page — force revealed (teaching) N/S/horizontal poles, current out of page — force withheld (assessment)
ART JOB 04	`unit202.fleming-left-hand.teaching` Fleming left-hand rule — motor teaching mnemonic	REQUIRED	READY	Wikimedia Commons — Fleming's Left Hand Rule.png	Fleming's left-hand mnemonic (teaching only)
ART JOB 05	`unit202.generator.rotating-loop.horizontal` Simple rotating-loop AC generator — loop plane facing poles — near-zero EMF	REQUIRED	READY	Wikimedia Commons — Diagram of single-phase generator with two poles.svg	Loop plane facing poles — near-zero EMF
ART JOB 06	`unit202.fleming-right-hand.teaching` Fleming right-hand rule — generator teaching mnemonic	REQUIRED	READY	Fleming's original 1902 right-hand-rule figure (Wikimedia Commons)	Fleming's right-hand mnemonic (teaching only)
ART JOB 07	`unit202.levers.class-1` Lever — Class I	REQUIRED	READY	Pearson Scott Foresman — Lever (PSF).svg	Class I recognition (no distances) Class I with effort-arm/load-arm distances
ART JOB 08	`unit202.levers.class-2` Lever — Class II	REQUIRED	READY	Pearson Scott Foresman — Lever (PSF).svg	Class II recognition (no distances) Class II with effort-arm/load-arm distances
ART JOB 09	`unit202.levers.class-3` Lever — Class III	REQUIRED	READY	Pearson Scott Foresman — Lever (PSF).svg	Class III recognition (no distances) Class III with effort-arm/load-arm distances
ART JOB 10	`unit202.pulleys.fixed` Fixed pulley	REQUIRED	READY	Wikimedia Commons — Pulley1a.svg	Fixed pulley

#	Asset	Need	Status	Reference	States supported
ART JOB 11	`unit202.pulleys.movable` Movable pulley	REQUIRED	READY	Wikimedia Commons — Pulley1a.svg	Movable pulley
ART JOB 12	`unit202.magnet.field` Bar magnet and magnetic field	REQUIRED	READY	Wikimedia Commons — DipolMagnet.svg	Bar magnet field pattern (no density comparison) Bar magnet field with flux-density comparison (same flux, different cross-section)
ART JOB 13	`unit202.motor.effect.vertical-poles` Motor effect — conductor in magnetic field (N/S poles arranged vertically)	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	N/S/vertical poles, current into page — force revealed (teaching) N/S/vertical poles, current into page — force withheld (assessment) N/S/vertical poles, current out of page — force revealed (teaching) N/S/vertical poles, current out of page — force withheld (assessment)
ART JOB 14	`unit202.generator.rotating-loop.vertical` Simple rotating-loop AC generator — loop plane edge-on to poles — near-peak EMF	REQUIRED	READY	Wikimedia Commons — Diagram of single-phase generator with two poles.svg	Loop plane edge-on to poles — near-peak EMF
ART JOB 15	`unit202.magnet.poles.like` Magnetic pole repulsion — like poles facing (repel)	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	like poles facing (repel) — force revealed (teaching) like poles facing (repel) — force withheld (assessment)
ART JOB 16	`unit202.gears.driven-larger` Driver/driven gears — driven gear larger than the driver	REQUIRED	READY	Wikimedia Commons — Example of a Compound Gear Train.png	Driver/driven gears — driven larger
ART JOB 17	`unit202.resistivity.length-comparison` Resistance vs conductor length	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Resistance vs conductor length
ART JOB 18	`unit202.resistivity.area-comparison` Resistance vs conductor cross-sectional area	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Resistance vs conductor cross-sectional area
ART JOB 19	`unit202.emf.motional` Motional EMF geometry	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Motional EMF — B, l, v mutually perpendicular
ART JOB 20	`unit202.components.physical.resistor` Physical electronic component — resistor	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — resistor
ART JOB 21	`unit202.components.physical.capacitor` Physical electronic component — capacitor	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — capacitor
ART JOB 22	`unit202.components.physical.diode` Physical electronic component — diode	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — diode
ART JOB 23	`unit202.components.physical.led` Physical electronic component — led	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — led

#	Asset	Need	Status	Reference	States supported
ART JOB 24	`unit202.components.physical.thermistor` Physical electronic component — thermistor	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — thermistor
ART JOB 25	`unit202.components.physical.transistor` Physical electronic component — transistor	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — transistor
ART JOB 26	`unit202.diode.bias-direction.forward` Diode forward bias — conducting	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Forward bias — conducting
ART JOB 27	`unit202.electrolysis` Chemical effect / electrolysis	REQUIRED	READY	Wikimedia Commons — Elektrolyse Allgemein.svg	Electrolysis cell — source, electrolyte, electrodes, current path
ART JOB 28	`unit202.magnet.poles.unlike` Magnetic pole attraction — unlike poles facing (attract)	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	unlike poles facing (attract) — force revealed (teaching) unlike poles facing (attract) — force withheld (assessment)
ART JOB 29	`unit202.gears.driven-smaller` Driver/driven gears — driven gear smaller than the driver	REQUIRED	READY	Wikimedia Commons — Example of a Compound Gear Train.png	Driver/driven gears — driven smaller
ART JOB 30	`unit202.gears.equal` Driver/driven gears — driver and driven gears equal size	REQUIRED	READY	Wikimedia Commons — Example of a Compound Gear Train.png	Driver/driven gears — equal
ART JOB 31	`unit202.diode.bias-direction.reverse` Diode reverse bias — blocked	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Reverse bias — blocked
ART JOB 32	`unit202.heating-effect` Heating effect of electric current	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Heating effect of electric current (blocked)
ART JOB 33	`unit202.conductor-insulator` Conductor vs insulator	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Conductor vs insulator (blocked)
ART JOB 34	`unit202.protective-devices` Fuse vs circuit breaker comparison	REQUIRED	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Fuse vs circuit breaker comparison (blocked)
ART JOB 35	`unit202.components.physical.zener-diode` Physical electronic component — zener diode	USEFUL	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — zener diode (blocked)
ART JOB 36	`unit202.components.physical.photodiode` Physical electronic component — photodiode	USEFUL	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — photodiode (blocked)
ART JOB 37	`unit202.components.physical.diac` Physical electronic component — DIAC	USEFUL	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — DIAC (blocked)
ART JOB 38	`unit202.components.physical.triac` Physical electronic component — TRIAC	USEFUL	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — TRIAC (blocked)
ART JOB 39	`unit202.components.physical.thyristor-scr` Physical electronic component — thyristor/SCR	USEFUL	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Physical appearance — thyristor/SCR (blocked)
ART JOB 40	`unit202.magnet.permanent-vs-electromagnet` Permanent magnet vs electromagnet comparison	USEFUL	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Permanent magnet vs electromagnet (blocked)

#	Asset	Need	Status	Reference	States supported
ART JOB 41	`unit202.instrument.clamp-meter` Clamp meter recognition	USEFUL	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Clamp meter recognition (blocked)
ART JOB 42	`unit202.instrument.oscilloscope` Oscilloscope recognition	USEFUL	BLOCKED	PRIMARY REFERENCE STILL TO BE APPROVED	Oscilloscope recognition (blocked)

Deterministic visual list

NO CHATGPT IMAGE GENERATION REQUIRED for any asset in this list -- all 11 are produced entirely by ALP's own rendering code.

Asset	Need	States
`unit202.circuit.series` Series circuit	REQUIRED	3
`unit202.circuit.parallel` Parallel circuit	REQUIRED	3
`unit202.circuit.mixed` Mixed series/parallel circuit	REQUIRED	2
`unit202.instrument.connections` Ammeter / voltmeter / ohmmeter connections	REQUIRED	5
`unit202.current-direction.electron-flow-vs-conventional` Conventional current vs electron flow	USEFUL	1
`unit202.gears.rotation-direction` Gear rotation-direction reversal / idler gear	USEFUL	2
`unit202.waveform.sine` AC sine waveform	REQUIRED	6
`unit202.components.symbols` Electronic component symbol system	REQUIRED	13
`unit202.rectification.waveforms` Rectifier/inverter output waveform shapes	REQUIRED	3
`unit202.capacitor.transient` Capacitor RC charge/discharge transient curve	REQUIRED	2
`unit202.trigonometry` Right-angle triangle / SOHCAHTOA	DEFERRED_SCOPE	1

Blocked references

REQUIRED, blocked (18)

Asset	Visual need	What reference is needed
`unit202.motor.effect.horizontal-poles` Motor effect — conductor in magnetic field (N/S poles arranged horizontally)	Show a current-carrying conductor between magnetic poles (N/S poles arranged horizontally) experiencing a force perpendicular to both the field and the current (the motor effect), distinct from the Fleming's-left-hand mnemonic itself.	An authoritative technical/photographic reference for: Motor effect — conductor in magnetic field (N/S poles arranged horizontally)
`unit202.motor.effect.vertical-poles` Motor effect — conductor in magnetic field (N/S poles arranged vertically)	Show a current-carrying conductor between magnetic poles (N/S poles arranged vertically) experiencing a force perpendicular to both the field and the current (the motor effect), distinct from the Fleming's-left-hand mnemonic itself.	An authoritative technical/photographic reference for: Motor effect — conductor in magnetic field (N/S poles arranged vertically)
`unit202.magnet.poles.like` Magnetic pole repulsion — like poles facing (repel)	Show like poles facing (repel) from the pole labels on two bar magnets.	An authoritative technical/photographic reference for: Magnetic pole repulsion — like poles facing (repel)
`unit202.magnet.poles.unlike` Magnetic pole attraction — unlike poles facing (attract)	Show unlike poles facing (attract) from the pole labels on two bar magnets.	An authoritative technical/photographic reference for: Magnetic pole attraction — unlike poles facing (attract)
`unit202.resistivity.length-comparison` Resistance vs conductor length	Show two conductor rods differing only in length so a learner predicts the qualitative effect on resistance.	An authoritative technical/photographic reference for: Resistance vs conductor length
`unit202.resistivity.area-comparison` Resistance vs conductor cross-sectional area	Show two conductor rods differing only in cross-sectional area so a learner predicts the qualitative effect on resistance.	An authoritative technical/photographic reference for: Resistance vs conductor cross-sectional area
`unit202.emf.motional` Motional EMF geometry	Show that conductor length, its velocity and the magnetic field are mutually perpendicular -- the geometric fact behind $e = Blv$.	An authoritative technical/photographic reference for: Motional EMF geometry
`unit202.components.physical.resistor` Physical electronic component — resistor	A physical-appearance companion image for the resistor, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.	An authoritative technical/photographic reference for: Physical electronic component — resistor
`unit202.components.physical.capacitor` Physical electronic component — capacitor	A physical-appearance companion image for the capacitor, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.	An authoritative technical/photographic reference for: Physical electronic component — capacitor
`unit202.components.physical.diode` Physical electronic component — diode	A physical-appearance companion image for the diode, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.	An authoritative technical/photographic reference for: Physical electronic component — diode
`unit202.components.physical.led` Physical electronic component — led	A physical-appearance companion image for the led, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.	An authoritative technical/photographic reference for: Physical electronic component — led

Asset	Visual need	What reference is needed
`unit202.components.physical.thermistor` Physical electronic component — thermistor	A physical-appearance companion image for the thermistor, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.	An authoritative technical/photographic reference for: Physical electronic component — thermistor
`unit202.components.physical.transistor` Physical electronic component — transistor	A physical-appearance companion image for the transistor, paired with its existing deterministic UK/IEC symbol card, so a learner can recognise this component both on a circuit diagram and in physical form.	An authoritative technical/photographic reference for: Physical electronic component — transistor
`unit202.diode.bias-direction.forward` Diode forward bias — conducting	CC-11.7 audit finding (new, beyond the original 66): show current flowing easily in forward bias, distinct from the static IEC symbol -- directly targets EL-COMPONENT-DIODE-001 and the named misconception MIS-EL-DIODE-DIRECTION-CONFUSION-001 (confusing which direction a diode conducts), which the existing `electronics.component_symbol_card` blueprint cannot represent since it renders only the static symbol, never current flow.	An authoritative technical/photographic reference for: Diode forward bias — conducting
`unit202.diode.bias-direction.reverse` Diode reverse bias — blocked	CC-11.7 audit finding (new, beyond the original 66): show current blocked in reverse bias, distinct from the static IEC symbol -- directly targets EL-COMPONENT-DIODE-001 and the named misconception MIS-EL-DIODE-DIRECTION-CONFUSION-001 (confusing which direction a diode conducts), which the existing `electronics.component_symbol_card` blueprint cannot represent since it renders only the static symbol, never current flow.	An authoritative technical/photographic reference for: Diode reverse bias — blocked
`unit202.heating-effect` Heating effect of electric current	Show the heating effect of electric current (resistive heating) at a conceptual level.	An authoritative technical/photographic reference for: Heating effect of electric current
`unit202.conductor-insulator` Conductor vs insulator	Show a material-recognition comparison between conductors and insulators.	An authoritative technical/photographic reference for: Conductor vs insulator
`unit202.protective-devices` Fuse vs circuit breaker comparison	Show a fuse and a circuit breaker side by side, supporting cap.fault.compare_fuse_breaker (fuse must be replaced once blown; breaker can be reset), without endorsing one manufacturer's product appearance as canonical.	An authoritative technical/photographic reference for: Fuse vs circuit breaker comparison

USEFUL, blocked (8)

Asset	Visual need	What reference is needed
`unit202.magnet.permanent-vs-electromagnet` Permanent magnet vs electromagnet comparison	CC-11.7 audit finding: a genuine physical-topology comparison (coiled wire around a core vs a solid bar magnet) the learner can use to distinguish the two magnet types by appearance -- USEFUL enrichment, Level 2 depth keeps it below REQUIRED.	An authoritative technical/photographic reference for: Permanent magnet vs electromagnet comparison
`unit202.instrument.clamp-meter` Clamp meter recognition	CC-11.7 audit finding: physical recognition of a clamp meter by its distinctive ferrite-jaw form -- genuinely different from the series/parallel/isolated connection topology the sibling TECHNICAL_DIAGRAM asset models. USEFUL enrichment, not REQUIRED.	An authoritative technical/photographic reference for: Clamp meter recognition
`unit202.instrument.oscilloscope` Oscilloscope recognition	CC-11.7 audit finding: physical recognition of an oscilloscope by its distinctive screen/trace form. USEFUL enrichment, not REQUIRED.	An authoritative technical/photographic reference for: Oscilloscope recognition
`unit202.components.physical.zener-diode` Physical electronic component — zener diode	A physical-appearance companion image for the zener diode, paired with its existing deterministic UK/IEC symbol card -- more specialist than the six REQUIRED components, genuinely useful but not REQUIRED for Unit 202 visual completeness (CC-11.7 audit finding).	An authoritative technical/photographic reference for: Physical electronic component — zener diode

Asset	Visual need	What reference is needed
`unit202.components.physical.photodiode` Physical electronic component — photodiode	A physical-appearance companion image for the photodiode, paired with its existing deterministic UK/IEC symbol card -- more specialist than the six REQUIRED components, genuinely useful but not REQUIRED for Unit 202 visual completeness (CC-11.7 audit finding).	An authoritative technical/photographic reference for: Physical electronic component — photodiode
`unit202.components.physical.diac` Physical electronic component — DIAC	A physical-appearance companion image for the DIAC, paired with its existing deterministic UK/IEC symbol card -- more specialist than the six REQUIRED components, genuinely useful but not REQUIRED for Unit 202 visual completeness (CC-11.7 audit finding).	An authoritative technical/photographic reference for: Physical electronic component — DIAC
`unit202.components.physical.triac` Physical electronic component — TRIAC	A physical-appearance companion image for the TRIAC, paired with its existing deterministic UK/IEC symbol card -- more specialist than the six REQUIRED components, genuinely useful but not REQUIRED for Unit 202 visual completeness (CC-11.7 audit finding).	An authoritative technical/photographic reference for: Physical electronic component — TRIAC
`unit202.components.physical.thyristor-scr` Physical electronic component — thyristor/SCR	A physical-appearance companion image for the thyristor/SCR, paired with its existing deterministic UK/IEC symbol card -- more specialist than the six REQUIRED components, genuinely useful but not REQUIRED for Unit 202 visual completeness (CC-11.7 audit finding).	An authoritative technical/photographic reference for: Physical electronic component — thyristor/SCR

Right-hand / Fleming / motor / generator directional safety

Allows independent technical review of every direction-sensitive electromagnetic family before any artwork is generated.

Right-hand grip rule / field around a conductor

Technical authority	Right-hand grip rule: thumb = conventional current, curled fingers = magnetic-field circulation direction. Governed by the deterministic magnetic.field_conductor_direction blueprint for assessment; the premium phenomenon/mnemonic assets are teaching-only illustrations of the same governed geometry.
Generated-art boundary	The conductor's concentric field-line pattern is direction-neutral in the base artwork -- current-direction and field-circulation indicators are added as a deterministic overlay, never baked in, so one base image safely serves both current directions (CC-11.7B correction).
Mirroring policy	**unit202.right-hand-grip.teaching: DO NOT MIRROR OR HORIZONTALLY FLIP HAND-RULE ARTWORK. Mirroring may change handedness (a mirrored right hand can read as a left hand) and invalidate the mnemonic -- never use a mirror/flip transform of this image to represent a reversed current direction.**
States requiring separate artwork	none
Deterministic states	none

Fleming's left-hand rule / motor effect

Technical authority	Fleming's left-hand rule (motor effect): thumb = force/motion, first finger = field, second finger = current, mutually perpendicular. Governed by the deterministic motor.force_field_current blueprint for assessment.
Generated-art boundary	Pole orientation (horizontal vs vertical) is a genuine apparatus-layout change requiring separate artwork (CC-11.7B split into .horizontal-poles / .vertical-poles); current direction within one fixed orientation remains a deterministic overlay concern.
Mirroring policy	**unit202.fleming-left-hand.teaching: DO NOT MIRROR OR HORIZONTALLY FLIP HAND-RULE ARTWORK. Mirroring may change handedness (a mirrored left hand can read as a right hand) and invalidate the mnemonic -- never use a mirror/flip transform of this image for any other pole/current combination.**
States requiring separate artwork	none
Deterministic states	none

Fleming's right-hand rule / AC generator effect

Technical authority	Fleming's right-hand rule (generator effect): thumb = conductor motion, first finger = field, second finger = induced current/EMF. Governed by the deterministic generator.rotating_loop blueprint for assessment.
Generated-art boundary	Loop pose (face-on/near-zero-EMF vs edge-on/near-peak-EMF) is a materially different 3D silhouette requiring separate artwork (CC-11.7B split into .horizontal / .vertical) -- no deterministic overlay can rotate a loop's rendered pose.
Mirroring policy	**unit202.fleming-right-hand.teaching: DO NOT MIRROR OR HORIZONTALLY FLIP HAND-RULE ARTWORK. Mirroring may change handedness (a mirrored right hand can read as a left hand) and invalidate the mnemonic -- never use a mirror/flip transform of this image for any other rotation phase.**
States requiring separate artwork	unit202.generator.rotating-loop.horizontal / unit202.generator.rotating-loop.horizontal.state.near-zero-emf unit202.generator.rotating-loop.vertical / unit202.generator.rotating-loop.vertical.state.near-peak-emf
Deterministic states	none

Component-recognition section

Every individual physical-recognition art job, shown separately -- never hidden behind one family count. 13 total.

Asset	Need	Status
`unit202.instrument.clamp-meter` Clamp meter recognition	USEFUL	BLOCKED
`unit202.instrument.oscilloscope` Oscilloscope recognition	USEFUL	BLOCKED
`unit202.components.physical.resistor` Physical electronic component — resistor	REQUIRED	BLOCKED
`unit202.components.physical.capacitor` Physical electronic component — capacitor	REQUIRED	BLOCKED
`unit202.components.physical.diode` Physical electronic component — diode	REQUIRED	BLOCKED
`unit202.components.physical.led` Physical electronic component — led	REQUIRED	BLOCKED
`unit202.components.physical.thermistor` Physical electronic component — thermistor	REQUIRED	BLOCKED
`unit202.components.physical.transistor` Physical electronic component — transistor	REQUIRED	BLOCKED
`unit202.components.physical.zener-diode` Physical electronic component — zener diode	USEFUL	BLOCKED
`unit202.components.physical.photodiode` Physical electronic component — photodiode	USEFUL	BLOCKED
`unit202.components.physical.diac` Physical electronic component — DIAC	USEFUL	BLOCKED
`unit202.components.physical.triac` Physical electronic component — TRIAC	USEFUL	BLOCKED
`unit202.components.physical.thyristor-scr` Physical electronic component — thyristor/SCR	USEFUL	BLOCKED

Final production readiness table

Asset	Need	Status
`unit202.current-conductor.magnetic-field` Magnetic field around a current-carrying conductor	REQUIRED	**READY TO GENERATE**
`unit202.right-hand-grip.teaching` Right-hand grip rule — teaching mnemonic	REQUIRED	**READY TO GENERATE**
`unit202.motor.effect.horizontal-poles` Motor effect — conductor in magnetic field (N/S poles arranged horizontally)	REQUIRED	**BLOCKED**
`unit202.motor.effect.vertical-poles` Motor effect — conductor in magnetic field (N/S poles arranged vertically)	REQUIRED	**BLOCKED**
`unit202.fleming-left-hand.teaching` Fleming left-hand rule — motor teaching mnemonic	REQUIRED	**READY TO GENERATE**
`unit202.generator.rotating-loop.horizontal` Simple rotating-loop AC generator — loop plane facing poles — near-zero EMF	REQUIRED	**READY TO GENERATE**
`unit202.generator.rotating-loop.vertical` Simple rotating-loop AC generator — loop plane edge-on to poles — near-peak EMF	REQUIRED	**READY TO GENERATE**
`unit202.fleming-right-hand.teaching` Fleming right-hand rule — generator teaching mnemonic	REQUIRED	**READY TO GENERATE**
`unit202.levers.class-1` Lever — Class I	REQUIRED	**READY TO GENERATE**
`unit202.levers.class-2` Lever — Class II	REQUIRED	**READY TO GENERATE**
`unit202.levers.class-3` Lever — Class III	REQUIRED	**READY TO GENERATE**
`unit202.pulleys.fixed` Fixed pulley	REQUIRED	**READY TO GENERATE**
`unit202.pulleys.movable` Movable pulley	REQUIRED	**READY TO GENERATE**
`unit202.magnet.field` Bar magnet and magnetic field	REQUIRED	**READY TO GENERATE**
`unit202.magnet.poles.like` Magnetic pole repulsion — like poles facing (repel)	REQUIRED	**BLOCKED**
`unit202.magnet.poles.unlike` Magnetic pole attraction — unlike poles facing (attract)	REQUIRED	**BLOCKED**
`unit202.magnet.permanent-vs-electromagnet` Permanent magnet vs electromagnet comparison	USEFUL	**BLOCKED**
`unit202.instrument.clamp-meter` Clamp meter recognition	USEFUL	**BLOCKED**
`unit202.instrument.oscilloscope` Oscilloscope recognition	USEFUL	**BLOCKED**
`unit202.gears.driven-larger` Driver/driven gears — driven gear larger than the driver	REQUIRED	**READY TO GENERATE**
`unit202.gears.driven-smaller` Driver/driven gears — driven gear smaller than the driver	REQUIRED	**READY TO GENERATE**
`unit202.gears.equal` Driver/driven gears — driver and driven gears equal size	REQUIRED	**READY TO GENERATE**
`unit202.resistivity.length-comparison` Resistance vs conductor length	REQUIRED	**BLOCKED**

Asset	Need	Status
`unit202.resistivity.area-comparison` Resistance vs conductor cross-sectional area	REQUIRED	BLOCKED
`unit202.emf.motional` Motional EMF geometry	REQUIRED	BLOCKED
`unit202.components.physical.resistor` Physical electronic component — resistor	REQUIRED	BLOCKED
`unit202.components.physical.capacitor` Physical electronic component — capacitor	REQUIRED	BLOCKED
`unit202.components.physical.diode` Physical electronic component — diode	REQUIRED	BLOCKED
`unit202.components.physical.led` Physical electronic component — led	REQUIRED	BLOCKED
`unit202.components.physical.thermistor` Physical electronic component — thermistor	REQUIRED	BLOCKED
`unit202.components.physical.transistor` Physical electronic component — transistor	REQUIRED	BLOCKED
`unit202.diode.bias-direction.forward` Diode forward bias — conducting	REQUIRED	BLOCKED
`unit202.diode.bias-direction.reverse` Diode reverse bias — blocked	REQUIRED	BLOCKED
`unit202.components.physical.zener-diode` Physical electronic component — zener diode	USEFUL	BLOCKED
`unit202.components.physical.photodiode` Physical electronic component — photodiode	USEFUL	BLOCKED
`unit202.components.physical.diac` Physical electronic component — DIAC	USEFUL	BLOCKED
`unit202.components.physical.triac` Physical electronic component — TRIAC	USEFUL	BLOCKED
`unit202.components.physical.thyristor-scr` Physical electronic component — thyristor/SCR	USEFUL	BLOCKED
`unit202.electrolysis` Chemical effect / electrolysis	REQUIRED	READY TO GENERATE
`unit202.heating-effect` Heating effect of electric current	REQUIRED	BLOCKED
`unit202.conductor-insulator` Conductor vs insulator	REQUIRED	BLOCKED
`unit202.protective-devices` Fuse vs circuit breaker comparison	REQUIRED	BLOCKED

Final audit statement

PASS — all mechanical audit gates are zero.

Historical variants unmapped	0 (target 0)
Duplicate ids	0 (target 0)
Premium/hybrid assets with no working prompt	0 (target 0)
Premium/hybrid assets with no reference or blocked status	0 (target 0)
Canonical states with no pedagogical state	0 (target 0)
Assessment states leaking a mnemonic	0 (target 0)
CC-11.7 USEFUL findings missing from catalogue	0 (target 0)
Catalogue structural problems	0 (target 0)
Multi-state assets missing a shared-base decision	0 (target 0)
Split decisions not actually applied	0 (target 0)
REQUIRED-but-blocked assets excluded from REQUIRED total	0 (target 0)

Run `npm run visuals:studio:audit` to reproduce this result mechanically.